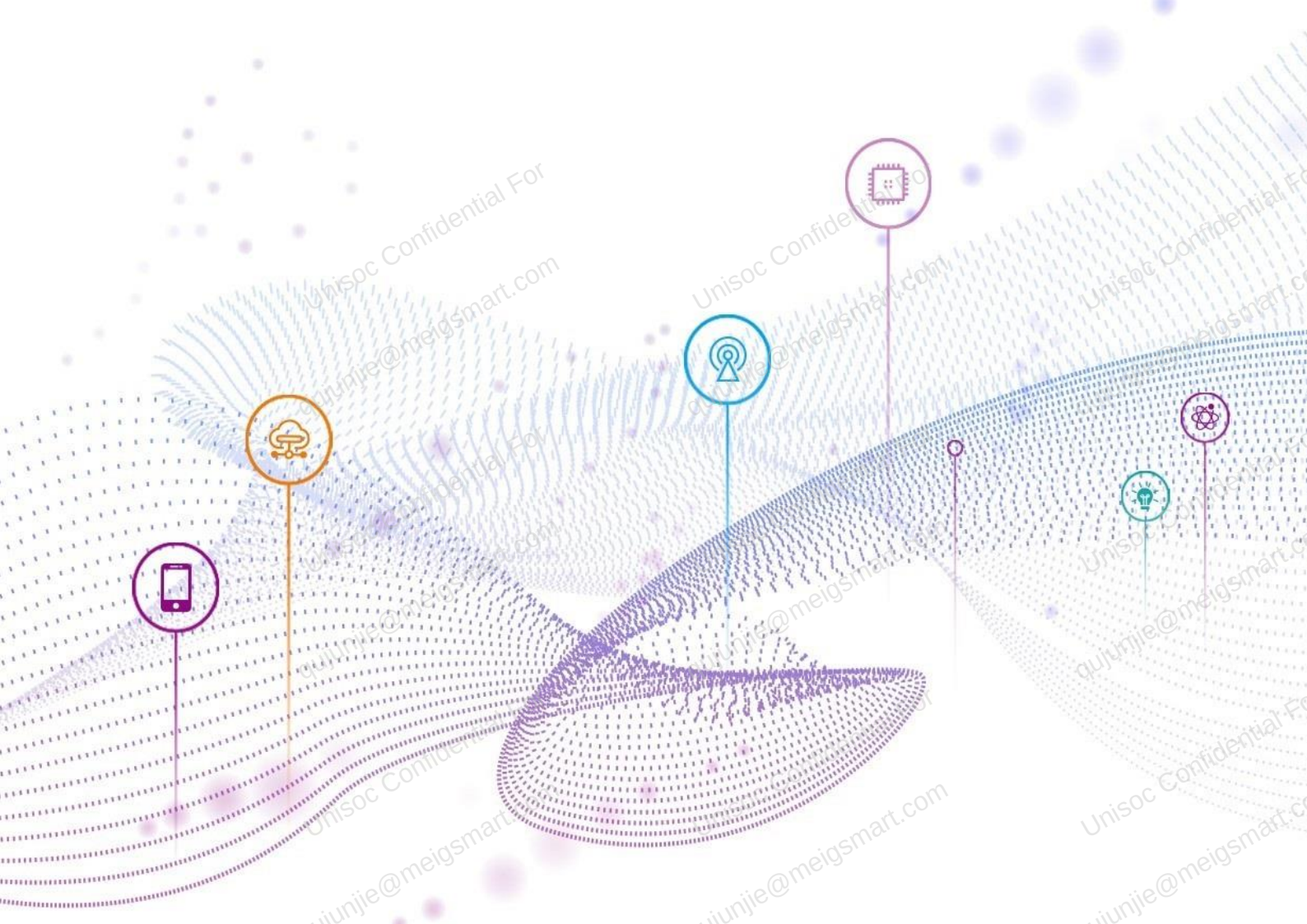




UIS7885 Brief Device Specification

Issue V1.8
Date 13-Jan-2026

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About This Document

Purpose

This document describes System Overview, Features, Package Information, Pin Information and Electrical Specification of UIS7885.

Intended Audience

This document is intended primarily for UIS7885 R&D and testing staff. Personnel must be familiar with the basic Knowledge of UNISOC Chip.

Acronyms and Abbreviations

Acronym and Abbreviation	Full Name
AC	Alternating Current
BGA	Ball Grid Array Package
DC	Direct Current
ESD	Electro Static discharge
PGA	Pin Grid Array Package
SIM	Subscriber Identity Module

Symbol Conventions

The symbols that may be found in this guide are defined in the following table.

Symbol	Description
 NOTE	Calls attention to important information, best practices and tips. NOTE is used to address information not related to personal injury, equipment damage, and environment deterioration.
 CAUTION	Calls attention to error-prone operations. CAUTION is used to address information not related to personal injury, equipment damage, and environment deterioration.

Symbol	Description
 WARNING	Calls attention to irreversible operations. WARNING is used to address information not related to personal injury and environment deterioration.

Change History

Issue	Date	Description
V1.8	13-Jan-2026	Update 3.5 Pin Multiplexed Function List, 3.6 Pin Status and GPIOs
V1.7	17-Jul-2025	Update Figure 1-1
V1.6	16-Jul-2025	Update 4.1.3 ESD Characteristics
V1.5	12-Jun-2025	- Update the voltage value of DCDC GEN0 - Update 1.4.3 Video Codec.
V1.4	22-May-2024	Update 1.4.3 Multimedia Features Camera feature.
V1.3	07-Jul-2023	Update Table 2-1 and Figure 2-3 .
V1.2	29-Jun-2023	Update 3.9.4 SPSEBC2_24X_VL/HL .
V1.1	06-Mar-2023	- Modify Modem's Specification. - Modify Uart mode with software/hardware flow control. - Optimize description of this issue. - Modify Video Codec's Specification - RFCTL 0~15 Renamed RFCTL 0(G0)~15 (G0) - RFCTL 16~31 Renamed RFCTL 0 (G1)~15 (G1)
V1.0	01-Aug-2022	This issue is the first official release.

Keywords

UIS7885, Datasheet, DeviceSpecification.

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1 System Overview

1.1 General Description

UIS7885 is a high performance application processor with integrated 2G/3G/4G/5G modem.

It consists of one-core Arm® Cortex®-A76 up to 2.7GHz, three-core Arm® Cortex®-A76 up to 2.3GHz, 4-core Arm® Cortex®-A55 up to 2.1GHz. In addition, an extensive set of interfaces is included to interface to cameras, touch panel display and UFS/eMMC/SD cards.

The application processor equipped with Arm® Neon™ engine support the latest Open OS along with its demanding applications such as web browsing, email and games.

The multi-standard multi-media accelerators and advanced audio subsystem are also integrated to provide advance multimedia applications and service such as streaming audio and video.

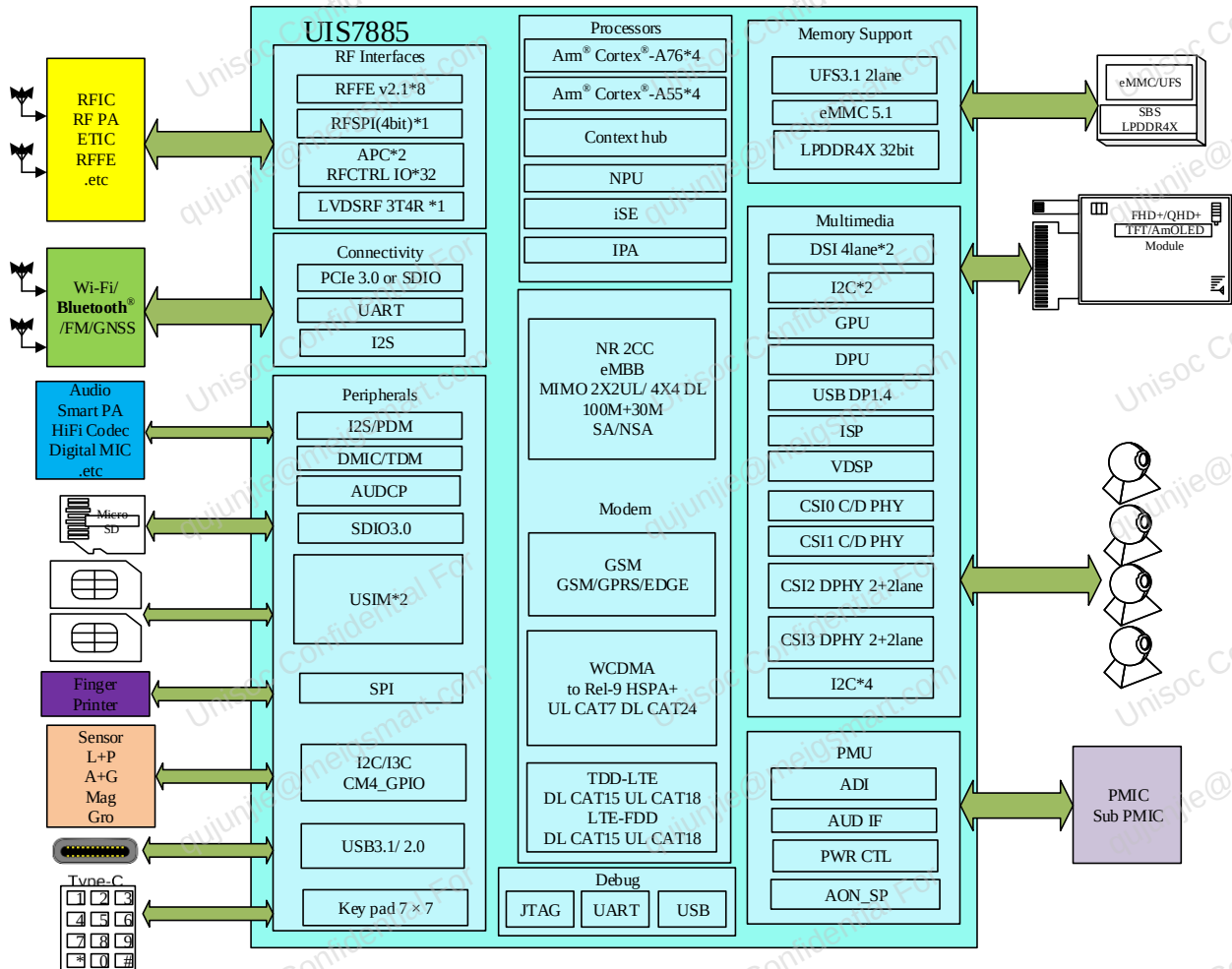
The high performance CPU and hardware accelerator are combined to provide modem subsystem capable of supporting NR FR1, LTE UL Cat18 and DL Cat15, Cat24 HSDPA and Cat7 HSUPA data rate, Class12 GPRS/EDGE.

The specially optimized architecture of UIS7885 can achieve high performance and low power for a lot of applications. Proprietary architectures and algorithms were developed for low power ASIC design and power management. Unique techniques are used for noise suppression/cancellation, echo suppression/cancellation algorithm. Overall, UIS7885 chipset presents a high performance-efficiency platform for 5G devices.

1.2 Functional Block Diagram

The following figure shows the chip-level functional block diagram of UIS7885.

Figure 1-1 Functional Block Diagram of UIS7885



1.3 Highlighted Features

- 6nm EUV process for best-in-class power-efficient 5G SOC chip
- One-core Arm® Cortex®-A76 up to 2.7GHz
- Three-core Arm® Cortex®-A76 up to 2.3GHz
- Four-core Arm® Cortex®-A55 up to 2.1GHz
- Shared 3MB L3 cache
- Arm® Mali™-G57, up to 850MHz
- IMG AX3596 up to 988MHz
- Support LPDDR4X, clock up to 2133MHz
- Support UFS3.1
- Support inter-band/Intra-band 2CC 100M + 30M
- FDD: DL Cat15 + UL Cat18(LTE only)
- TDD: DL Cat15 + UL Cat18(LTE only)

- EUTRA: DL Cat15, UL Cat18(EN-DC)
- Support QHD+@60fps/FHD+@120fps
- VP9: encoder to 1080p@30fps, decoder to profile0/profile2, 4k@30fps
- Single Camera up to 108M(9in1)
- Support Low Power Voice trigger

1.4 Features

1.4.1 Platform Features

AP subsystem

- One-core Arm® Cortex®-A76 up to 2.7GHz and three-core Arm® Cortex®-A76 up to 2.3GHz, each core with 64KB L1 I-cache, 64KB L1 D-cache and 256KB L2 cache
- Four-core Arm® Cortex®-A55 up to 2.1GHz, each core with 32KB L1 I-cache, 32KB L1 D-cache and 128KB L2 cache
- Shared 3MB L3 cache
- Arm® Neon™ multimedia processing engine with SIMDv2/VFPv4 ISA
- DVFS technology with operating voltage from 0.65V~0.95V

NPU Subsystem

- IMG AX3596 up to 988MHz
- Support a range of neural network operation such convolutions, activations, element-wise operations, pooling etc.
- Lossless weight data compression
- DRM security and scheduling for multiple hosts

VDSP Subsystem

VQ7 up to 1014MHz, 256KB Data RAM

Security Subsystem

Isolated execution environment

External memory interface

- Support eMMC 5.1 HS400
- Support LPDDR4X
- Support UFS3.1
- Supporting the following memory combinations:
 - eMCP(eMMC/LPDDR4X)
 - eMMC + LPDDR4X
 - uMCP(UFS3.1/LPDDR4X)
 - UFS3.1 + LPDDR4X
- eMMC flash is supported by eMMC controller, with the following features:

- Support 1.8V I/O
- Support 1.8 or 3.0V VCC
- Support 1/4/8-bit device
- Compliance to eMMC5.1
- LPDDR4X is supported by DDR Controller, with the features list below:
 - Support 2 × 16-bit LPDDR4X
 - Support dual channel
 - Support up to 32GB density
 - Support up to 2 CS
 - Support up to 32-bits data width
 - Clock up to 2133MHz
- UFS3.1 MPHY Support the following feature:
 - Support MPHY Type 1 M-ports
 - Internal(Through SOC) and external(Through PAD) reference clock supported: 26MHz
 - 2 TX lane and 2 RX lane operation
 - High Speed G1/2/3/4, Mode A/B operation
 - Support 1.8V and 1.2V Analog Supply
 - Compliance to MPHY V4.1
 - Compliance to UniPro V1.8

Peripheral and Connectivity Interfaces

- Support 2-SIM cards, both 1.8V and 3.0V devices
- Support 3 SDIO 3.0 with 4bit data width
- Support USB 2.0/3.1 Gen2/ USB 3.0 OTG
- Support 8 UARTs, 2 UARTs support hardware flow control
- Support 4 SPIs, support both master and slave
- Support 6 I2S(PCM) for audio codec connection
- Support 3 × 3 keypad default up to 7 × 7
- Support 10 I2Cs up to 3.4Mbps, include 2 I3Cs
- Support more than 198 GPIO pins
- Support 3 PWM outputs
- Support SIM/USIM
- Support 6 digital MICs
- Support 1 PCIE 3.0 port with 1 lane

1.4.2 Modem Features

NR Baseband

- 3GPP Release 15
- Support NSA/ SA
- Support NSA NR 1CC + LTE 1CC/ NR 1CC + LTE 2CC
- Support inter-band/Intra-band 2CC 100M + 30M

- Support 4×4 MIMO
- Support 256 QAM for DL and UL
- Support Cryptographic and integrity algorithms: NIA0/ NIA1/ NIA2/ NIA3 and NEA0/ NEA1/ NEA2/ NEA3
- Support PWS on NR
- Support Network slice
- Support VoNR and EPS fallback

TD-LTE/LTE FDD Baseband

- 3GPP Release 15
- FDD: DL Cat15 + UL Cat18 (LTE only)
- TDD: DL Cat15 + UL Cat18 (LTE only)
- Support DL $2/4$ layers MIMO($2 \times 2 / 4 \times 2 / 8 \times 2 / 4 \times 4 / 8 \times 4$)
- Support 256QAM for UL and DL
- Support intra/inter-band 2CA/3CA DL
- Support intra-band continuous 2CA UL
- Support FDD + TDD CA
- Support VoLTE/ViLTE
- Support MFBI
- Support Short DRX/ Long DRX/ C-DRX
- Support TTI Bundling/SPS
- Cryptographic and integrity algorithms: EEA1/EIA1:SNOW-3G, EEA2/EIA2:AES, EEA3/EIA3:ZUC
- Support RoHC
- Support PWS(Public Warning System, including ETWS and CMAS) receiving
- Support EN-DC

WCDMA/HSDPA/HSUPA Baseband

- 3GPP release 9
- Support downlink Category 24 (64QAM, dual-cell HSDPA), throughput up to 42Mbps
- Support uplink Category 7 (16QAM, HSUPA), throughput up to 11Mbps
- Support CPC/enhanced cell-FACH/enhanced F-DPCH/eDRX
- Support SRBoHS, CELL & URA-PCH
- Support DL diversity and type3i receiver
- Support Fast Dormancy
- Cryptographic and integrity algorithms: UEA1/UIA1: KASUMI, UEA2/UIA2: SNOW-3G
- Support PWS(Public Warning System, including ETWS and CMAS)receiving

GSM/GPRS/EDGE Baseband

- GSM/GPRS/EDGE Release 6, Release 9 particularly for LTE inter-operation
- Support GSM850, GSM900, DCS1800, and PCS1900
- Support GPRS/EDGE class12
- Support GPRS service: class B

- Cryptographic algorithms: A5/1 A5/3, GEA1 GEA2 GEA3
- Support DL diversity receiver
- Support VAMOS I
- Support DARPI
- Support PWS(Public Warning System, including ETWS and CMAS) receiving

Multi-mode

- Support DSDS
- Support eSRVCC/aSRVCC/bSRVCC/mid-SRVCC
- Support PSHO/ CSFB/ Redirection

Voice Codec

- Support quad vocoders for AMR-NB, EFR, HR and FR
- Support AMR-WB/AMR-NB
- Support EVS-NB/WB/SWB*
- Support noise reduction
- Support echo suppression/echo cancellation

1.4.3 Multimedia Features

3D Graphics

- Arm® Mali™-G57 4cores, up to 850MHz
- Support OpenGL ES 3.2/3.1/3.0/2.0/1.1
- OpenCL 2.x
- Vulkan 1.1
- Support DVFS technology

Image(JPEG)

- Compliance with JPEGISO/IEC 10918-1
- Support encoder: YUV420/422
- Support decoder: YUV420/422/400
- Support 640M pixels/s, 10 fps continuous shoot

Display

- Support DSI v1.2 and D-PHY v1.2
- Support Dual MIPI
- Support QHD+@60fps/FHD+@120fps
- Support extend panel by DP V1.4
- Support 8 layer alpha-blending and writeback
- Support layer rotation
- Support color space conversion
- Support ProtectView:

- Support Blue Light Protection
- Support Sun Light Protection
- Support ViveView:
 - Support contrast enhancement
 - Support sharpness enhancement
 - Support HDR Playback
 - Support Wide Gamut Display, sRGB/DCI-P3
- Support GreenView:
 - Ambient Adaptive Lightness
 - Content Adaptive Brightness Control
 - Super Resolution

Camera

- Support four camera
 - Single Camera: 64M(4in1) 25fps ZSL capture, 108M(9in1) NonZSL capture
 - Dual Camera: 32M 25fps + 8M 25fps, 20M + 20M, 24M + 16M
 - Triple Camera: 64M(4in1) + 20M(UW) + 13M(T)
 - Four Camera: 64M(4in1) + 20M(UW) + 13M(T) + 2M
- Support four port MIPI CSI-2
 - MIPI CSI port0, 4-datalane, CPHY and DPHY combo, 3.5Gbps/Trio, 2.5Gbps/lane
 - MIPI CSI port1, 4-datalane, CPHY and DPHY combo, 2.5Gbps/Trio, 2.5Gbps/lane
 - MIPI CSI port2, 4-datalane or 2×2-datalane, 2.5Gbps/lane, support TX for log
 - MIPI CSI port3, 4-datalane or 2×2-datalane, 2.5Gbps/lane, support TX for log
- Support AI Face Detection
- Support dual PDAF
- Support TOF
- Support slow motion, up to 480fps@720p & 240fps@1080p, 60fps@4K
- Support super low light capturing for still and video
- Support auto scene recognition by SW or VDSP clarification:
 - Portrait
 - Children
 - Backlight & portrait
 - Night
 - Night & portrait
 - Low light
 - Document
 - Macro
 - Standard
 - Landscape
 - Black and white
 - Spotlight
 - Sports

- HDR
- Support Face recognition

Video Codec

- H.264
 - Compliance with ISO/IEC 14496-10-MPEG4 Part10, Advanced Video Coding & Scalable Video Coding
 - Encoder: progressive mode, YUV420, 8-bits
 - Decoder: progressive & interlaced mode, YUV420, 8-bits/10-bits
 - Encoder: baseline profile, main profile, high profile, level 5.1, 4k@24fps, 1080p@100fps, 720p@240fps
 - Decoder: baseline profile, main profile, high profile, high 10 profile, level 5.1, 4k@30fps
- H.265
 - Compliance with ISO/IEC 23008-2-Part2: High efficiency video coding
 - Encoder: Progressive mode, YUV420, 8-bits
 - Decoder: progressive mode, YUV420, 8-bits/10-bits
 - Encoder: main profile, level 5.1, 4k@60fps, 1080p@240fps
 - Decoder: main profile, main 10 profile, level 5.1, 4k@30fps 10bit, 4k@60fps 8bit
- VP8
 - Encoder to 1080p@30fps by software
 - Decoder to 1080p@60fps
- VP9
 - Encoder to 1080p@30fps by software
 - Decoder: profile0, profile2, 4k@30fps

FBC

- Support frame buffer compression
- Support AFBC1.2

Audio

- Support MP3/AAC/ADPCM/AMR/AMR_WB/FLAC decoding
- Support PCM record and playback
- Support Audio processing algorithm
- Support lower power MP3 decoding
- Support lower power A2DP
- Support USB audio
- Support low power Voice trigger

2 Package Information

2.1 Basic Information

Plastic-encapsulated surface mount packages are sensitive to damage induced by absorbed moisture and temperature. The chip is MSL 3, which is marked on the label of the package.

Table 2-1 Package Information

Device Name	UIS7885
Type	FCCSP
Body Size	12.5mm × 12.0mm
Height	0.869mm maximum
Ball count	852
Ball Pitch	0.4mm
Ball Size	0.26mm
Ball Matrix	43 × 41

2.2 Top Marking Definition

Figure 2-1 Silk screen printing

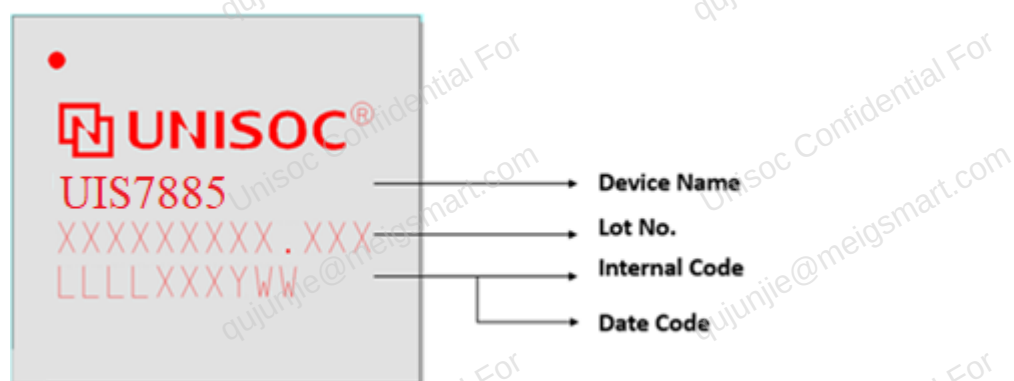


Figure 2-2 Package outline



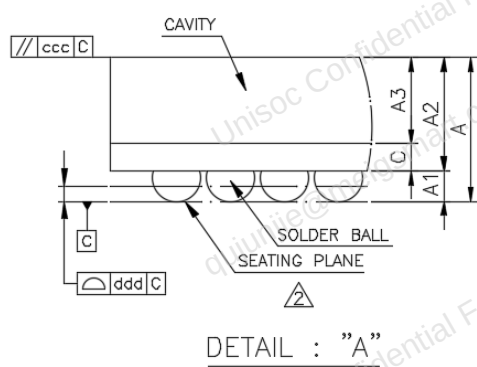
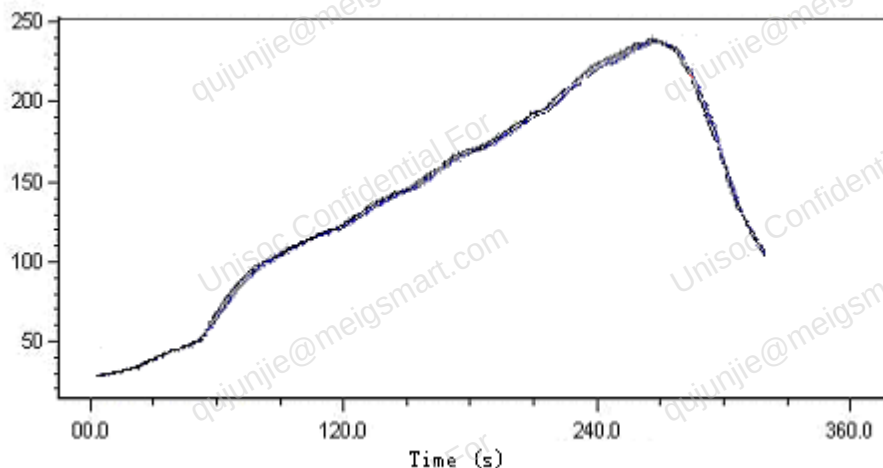


Figure 2-3 Package parameters

Symbol	Dimension in mm			Dimension in inch		
	MIN	NOM	MAX	MIN	NOM	MAX
A	0.7390	0.8040	0.8690	0.0291	0.0317	0.0342
A1	0.1200	0.1700	0.2200	0.0047	0.0067	0.0087
A2	0.5920	0.6340	0.6760	0.0233	0.0250	0.0266
A3	0.4200	0.4500	0.4800	0.0165	0.0177	0.0189
c	0.1540	0.1840	0.2140	0.0061	0.0072	0.0084
D	12.4000	12.5000	12.6000	0.4882	0.4921	0.4961
E	11.9000	12.0000	12.1000	0.4685	0.4724	0.4764
D1	——	11.8797	——	——	0.4677	——
E1	——	11.3140	——	——	0.4454	——
e	——	0.4000	——	——	0.0157	——
b	0.2100	0.2600	0.3100	0.0083	0.0102	0.0122
aaa	0.1000			0.0039		
ccc	0.2000			0.0079		
ddd	0.0800			0.0031		
eee	0.1500			0.0059		
fff	0.0500			0.0020		
MD/ME	43/41					
Note: refer to the SMT capacity for the package dimension of b.						

2.4 Reflow Profile

Figure 2-4 Reflow profile



NOTE

- Recommended reflow profile for lead-free solder paste
 - Ramp at 1-2°C per second to 245±5°C
 - Dwell at 235°C for 10s
 - Dwell at 217°C for 30~60s
 - Total reflow time is about 220~270s
 - Cold down ramp <4°C/s
- Recommended solder paste type
 - SnAgCu solder paste
 - Metal contents should be about 88.5%
- Recommended parameter for stencil making
 - Metal mask thickness: 5 mils
 - Opening area ratio: 100%

2.5 Device Moisture Sensitivity Level

Plastic-encapsulated surface mount packages are sensitive to damage induced by absorbed moisture and temperature. UIS7885 is MSL 3.

Table 2-2 MSL ratings summary

MSL	Out-of-bag floor life	Comments
1	Unlimited	<=30°C/85% RH
2	1 year	<=30°C/60% RH

MSL	Out-of-bag floor life	Comments
2a	4 weeks	<=30°C/60% RH
3	168 hours	<=30°C/60% RH
4	72 hours	<=30°C/60% RH
5	48 hours	<=30°C/60% RH
5a	24 hours	<=30°C/60% RH
6	Mandatory bake before use. After bake, must be reflowed within the time limit specified on the label.	<=30°C/60% RH

2.6 Thermal Characteristics

The thermal characteristics are as shown in [Table 2-3](#).

Table 2-3 Thermal characteristics

Symbol	Parameter	Condition	Value	Unit
Theta JA	Junction-to-Ambient thermal resistance	Air flow: 0 m/s	25.2	°C/W

3 Pin Definition

3.1 BGA Pin out

Figure 3-1 BGA Pin out



3.2 Pin Symbol Description

Table 3-1 Definition of pin symbols

Field	Symbol	Type Description
PinType	I	Digital input
	O	Digital output

Field	Symbol	Type Description
	O/T	Digital output with tri-state option
	I/O	Digital bi-directional pin
	I/O/T	Digital bi-directional pin with tri-state option
	PI	Power pin, input from external power supply
	PIO	Power pin, input from external or floating to use internal LDO power supply
	PO	Power pin, output for external devices
	G	Ground pin
	AI	Analog input pin
	AO	Analog output pin
	AIO	Analog bi-directional pin
Pin Value	IPU	Input with pull-up
	IPD	Input with pull-down
	OH	Output “1”
	OL	Output “0”
	Hiz	Tri-state
Power	VBAT	Battery power supply input
	VDDCORE	Power supply for core
	VDDCPU_LIT	Power supply for Little core CPU
	VDDCPU_MID	Power supply for Middle core CPU
	VDDCPU_BIG	Power supply for Big core CPU
	VDDGPU	Power supply for GPU
	VDDMM	Power supply for Multi-Media
	VDDMODEM	Power supply for Modem
	VDDPUB	Power supply for DDR controller
	VDDSRAM	Power supply for Dual-rail SRAM
	VDDSRAM_CPU_LIT	Power supply for Dual-rail SRAM of Little core CPU
	VDDSRAM_CPU_MID	Power supply for Dual-rail SRAM of Middle core CPU
	VDDAI	Power supply for NPU
	VSD0	Power supply for SDIO0 IO
	VSD2	Power supply for SDIO2 IO
	VEMMCIO	Power supply for eMMC IO

Field	Symbol	Type Description
	VSIM0 VSIM1	Power supply for SIM card 0/1
	VIO1V8	Power supply for I/O
	VIO1V8_ISE	ISE 1.8V digital power supply
	VDDMEM	Power supply for LPDDR4X core power
	VDDMEMQ	Power supply for LPDDR4X IO
	VDD0V75_PCIE	PCIe 0.75V digital power supply
	VDD0V75_UFS	UFS 0.75V digital power supply
	VDD0V75_USB	USB3.1 0.75V digital power supply
	AVDD1V2_DSI	DSI 1.2V analog power supply
	AVDD1V2_LVDSRF	LVDSRF 1.2V analog power supply
	AVDD1V2_CSI	CSI 1.2V analog power supply
	AVDD1V2_PCIE	PCIe 1.2V analog power supply
	AVDD1V2_UFS	UFS 1.2V analog power supply
	AVDD1V2_USB	USB3.1 1.2V analog power supply
	AVDD1V8_DPLL	DDR PLL 1.8V analog power supply
	AVDD1V8_DSI	DSI 1.8V analog power supply
	AVDD1V8_USB	USB2.0 1.8V analog power supply
	AVDD1V8_MPLLBIG	MPLLBIG 1.8V analog power supply
	AVDD1V8_MPLLLIT	MPLLLIT 1.8V analog power supply
	AVDD1V8_MPLLMID	MPLLMID 1.8V analog power supply
	AVDD1V8_PCIEPLL	PCIEPLL 1.8V analog power supply
	AVDD1V8_USBPLL	USB3.1 PLL 1.8V analog power supply
	AVDD1V8_EFU	EFUSE 1.8V analog power supply
	AVDD1V8_BB	Baseband DAC & PLL 1.8V analog power supply
	AVDD1V8_LVDSRF	LVDSRF 1.8V analog power supply
	AVDD1V8_CSI	CSI 1.8V analog power supply
	AVDD3V3_USB	USB2.0 3.3V analog power supply
Pin description	ie	Input enable, ie=0 means input disable, ie=1 means input enable
	oe	Output enable, oe=0 means output disable, oe=1, 1 means output enable, output high; oe=1, 0 means output enable, output low
	ie=0, oe=0	Hiz

Field	Symbol	Type Description
	wpu	Weak pull up resistor
	wpus	Weak pull up resistor selection
	wpd	Weak pull down resistor

3.3 Pin List

See *UIS7885 Pin List and Pin Register Instructions*.

3.4 Pin Description

Table 3-2 Pin description

Ball No	Ball Name	Description
LPDDR4X Interface		
N43	EMCA0_A/EMCA1_A	LPDDR4X channel A command or address 0/Discrete LPDDR4X channel A command or address 1
M42	EMCA1_A/EMCA0_A	LPDDR4X channel A command or address 1/Discrete LPDDR4X channel A command or address 0
K40	EMCA2_A/EMCA5_A	LPDDR4X channel A command or address 2/Discrete LPDDR4X channel A command or address 5
K42	EMCA3_A/EMCA3_A	LPDDR4X channel A command or address 3/Discrete LPDDR4X channel A command or address 3
L43	EMCA4_A/EMCA2_A	LPDDR4X channel A command or address 4/Discrete LPDDR4X channel A command or address 2
K38	EMCA5_A/EMCA4_A	LPDDR4X channel A command or address 5/Discrete LPDDR4X channel A command or address 4
M38	EMCK_C_A/EMCLK_A_N	LPDDR4X channel A differential clock negative/Discrete LPDDR4X channel A differential clock negative
N39	EMCK_T_A/EMCLK_A_P	LPDDR4X channel A differential clock positive/Discrete LPDDR4X channel A differential clock positive
L41	EMCKE0_A/EMCKE0_A	LPDDR4X channel A clock enable output 0/Discrete LPDDR4X channel A clock enable output 0
L39	EMCKE1_A/EMCKE1_A	LPDDR4X channel A clock enable output 1/Discrete LPDDR4X channel A clock enable output 1
M40	EMCS0_A/EMCS1_A	LPDDR4X channel A chip select 0/Discrete LPDDR4X channel A chip select 1

Ball No	Ball Name	Description
N41	EMCS1_A/EMCS0_A	LPDDR4X channel A chip select 1/Discrete LPDDR4X channel A chip select 0
H38	EMDMI0_A/EMDQ4_A	LPDDR4X channel A data mask Byte 0/Discrete LPDDR4X channel A data 4
E41	EMDQ0_A/EMDQ6_A	LPDDR4X channel A data 0/Discrete LPDDR4X channel A data 6
F40	EMDQ1_A/EMDQ7_A	LPDDR4X channel A data 1/Discrete LPDDR4X channel A data 7
G41	EMDQ2_A/EMDMI0_A	LPDDR4X channel A data 2/Discrete LPDDR4X channel A data mask Byte 0
H40	EMDQ3_A/EMDQ5_A	LPDDR4X channel A data 3/Discrete LPDDR4X channel A data 5
E43	EMDQ4_A/EMDQ0_A	LPDDR4X channel A data 4/Discrete LPDDR4X channel A data 0
G43	EMDQ5_A/EMDQ2_A	LPDDR4X channel A data 5/Discrete LPDDR4X channel A data 2
H42	EMDQ6_A/EMDQ3_A	LPDDR4X channel A data 6/Discrete LPDDR4X channel A data 3
F42	EMDQ7_A/EMDQ1_A	LPDDR4X channel A data 7/Discrete LPDDR4X channel A data 1
G39	EMDQS0_C_A/EMDQS0_A_N	LPDDR4X channel A data strobe 0 differential negative/Discrete LPDDR4X channel A data strobe 0 differential negative
F38	EMDQS0_T_A/EMDQS0_A_P	LPDDR4X channel A data strobe 0 differential positive/Discrete LPDDR4X channel A data strobe 0 differential positive
T40	EMDMI1_A/EMDQ11_A	LPDDR4X channel A data mask Byte 1/Discrete LPDDR4X channel A data 11
V38	EMDQ8_A/EMDQ8_A	LPDDR4X channel A data 8/Discrete LPDDR4X channel A data 8
V42	EMDQ9_A/EMDQ12_A	LPDDR4X channel A data 9/Discrete LPDDR4X channel A data 12
V40	EMDQ10_A/EMDQ9_A	LPDDR4X channel A data 10/Discrete LPDDR4X channel A data 9
U41	EMDQ11_A/EMDQ10_A	LPDDR4X channel A data 11/Discrete LPDDR4X channel A data 10
U43	EMDQ12_A/EMDQ13_A	LPDDR4X channel A data 12/Discrete LPDDR4X channel A data 13
R41	EMDQ13_A/EMDMI1_A	LPDDR4X channel A data 13/Discrete LPDDR4X channel A data mask Byte 1

Ball No	Ball Name	Description
T42	EMDQ14_A/EMDQ14_A	LPDDR4X channel A data 14/Discrete LPDDR4X channel A data 14
R43	EMDQ15_A/EMDQ15_A	LPDDR4X channel A data 15/Discrete LPDDR4X channel A data 15
T38	EMDQS1_C_A/EMDQS1_A_N	LPDDR4X channel A data strobe 1 differential negative/Discrete LPDDR4X channel A data strobe 1 differential negative
U39	EMDQS1_T_A/EMDQS1_A_P	LPDDR4X channel A data strobe 1 differential positive/Discrete LPDDR4X channel A data strobe 1 differential positive
AE43	EMCA0_B/EMCA1_B	LPDDR4X channel B command or address 0/Discrete LPDDR4X channel B command or address 1
AF42	EMCA1_B/EMCA0_B	LPDDR4X channel B command or address 1/Discrete LPDDR4X channel B command or address 0
AH40	EMCA2_B/EMCA5_B	LPDDR4X channel B command or address 2/Discrete LPDDR4X channel B command or address 5
AH42	EMCA3_B/EMCA3_B	LPDDR4X channel B command or address 3/Discrete LPDDR4X channel B command or address 3
AG43	EMCA4_B/EMCA2_B	LPDDR4X channel B command or address 4/Discrete LPDDR4X channel B command or address 2
AH38	EMCA5_B/EMCA4_B	LPDDR4X channel B command or address 5/Discrete LPDDR4X channel B command or address 4
AF38	EMCK_C_B/EMCLK_B_N	LPDDR4X channel B differential clock negative/Discrete LPDDR4X channel B differential clock negative
AE39	EMCK_T_B/EMCLK_B_P	LPDDR4X channel B differential clock positive/Discrete LPDDR4X channel B differential clock positive
AG41	EMCKE0_B/EMCKE0_B	LPDDR4X channel B clock enable output 0/Discrete LPDDR4X channel B clock enable output 0
AG39	EMCKE1_B/EMCKE1_B	LPDDR4X channel B clock enable output 1/Discrete LPDDR4X channel B clock enable output 1
AF40	EMCS0_B/EMCS1_B	LPDDR4X channel B chip select 0/Discrete LPDDR4X channel B chip select 1
AE41	EMCS1_B/EMCS0_B	LPDDR4X channel B chip select 1/Discrete LPDDR4X channel B chip select 0
AK38	EMDMI0_B/EMDQ4_B	LPDDR4X channel B data mask Byte 0/Discrete LPDDR4X channel B data 4
AN41	EMDQ0_B/EMDQ6_B	LPDDR4X channel B data 0/Discrete LPDDR4X channel B data 6
AM40	EMDQ1_B/EMDQ7_B	LPDDR4X channel B data 1/Discrete LPDDR4X channel B data 7

Ball No	Ball Name	Description
AL41	EMDQ2_B/EMDMI0_B	LPDDR4X channel B data 2/Discrete LPDDR4X channel B data mask Byte 0
AK40	EMDQ3_B/EMDQ5_B	LPDDR4X channel B data 3/Discrete LPDDR4X channel B data 5
AN43	EMDQ4_B/EMDQ0_B	LPDDR4X channel B data 4/Discrete LPDDR4X channel B data 0
AL43	EMDQ5_B/EMDQ2_B	LPDDR4X channel B data 5/Discrete LPDDR4X channel B data 2
AK42	EMDQ6_B/EMDQ3_B	LPDDR4X channel B data 6/Discrete LPDDR4X channel B data 3
AM42	EMDQ7_B/EMDQ1_B	LPDDR4X channel B data 7/Discrete LPDDR4X channel B data 1
AL39	EMDQS0_C_B/EMDQS0_B_N	LPDDR4X channel A data strobe 0 differential negative/Discrete LPDDR4X channel B data strobe 0 differential negative
AM38	EMDQS0_T_B/EMDQS0_B_P	LPDDR4X channel A data strobe 0 differential positive/Discrete LPDDR4X channel B data strobe 0 differential positive
AB40	EMDMI1_B/EMDQ11_B	LPDDR4X channel B data mask Byte 1/Discrete LPDDR4X channel B data 11
Y38	EMDQ8_B/EMDQ8_B	LPDDR4X channel B data 8/Discrete LPDDR4X channel B data 8
Y42	EMDQ9_B/EMDQ12_B	LPDDR4X channel B data 9/Discrete LPDDR4X channel B data 12
Y40	EMDQ10_B/EMDQ9_B	LPDDR4X channel B data 10/Discrete LPDDR4X channel B data 9
AA41	EMDQ11_B/EMDQ10_B	LPDDR4X channel B data 11/Discrete LPDDR4X channel B data 10
AA43	EMDQ12_B/EMDQ13_B	LPDDR4X channel B data 12/Discrete LPDDR4X channel B data 13
AC41	EMDQ13_B/EMDMI1_B	LPDDR4X channel B data 13/Discrete LPDDR4X channel B data mask Byte 1
AB42	EMDQ14_B/EMDQ14_B	LPDDR4X channel B data 14/Discrete LPDDR4X channel B data 14
AC43	EMDQ15_B/EMDQ15_B	LPDDR4X channel B data 15/Discrete LPDDR4X channel B data 15
AB38	EMDQS1_C_B/EMDQS1_B_N	LPDDR4X channel A data strobe 1 differential negative/Discrete LPDDR4X channel B data strobe 1 differential negative

Ball No	Ball Name	Description
AA39	EMDQS1_T_B/EMDQS1_B_P	LPDDR4X channel A data strobe 1 differential positive/Discrete LPDDR4X channel B data strobe 1 differential positive
W43	EMRSTN	LPDDR4X reset , low active
C43	ZQ	LPDDR4X Impedance calibration
Thermal Test Interface		
AU33	SSUSB_CKN	USB3.1 backup clock input negative
AW33	SSUSB_CKP	USB3.1 backup clock input positive
AT38	UFS_CKN	UFS backup clock input negative
AV38	UFS_CKP	UFS backup clock input positive
V10	CLKOUT_TEST_PLL	PLL clock test point
H26	CLKTEST_MPLLBIG	MPLLBIG clock test point
Y22	CLKTEST_MPLLLIT	MPLLLIT clock test point
Y8	TEST_THM_DP	Thermal Test Diode anode
F32	TSET_DDC_LS	DDC low speed test point
CSI Interface		
C5	CSI0_CKN_C1	CSI0 D-PHY differential clock 0 negative/CSI0 C-PHY lane 1 C
E5	CSI0_CKP_B1	CSI0 D-PHY differential clock 0 positive/CSI0 C-PHY lane 1 B
B2	CSI0_DN0_B0	CSI0 D-PHY differential data 0 negative/CSI0 C-PHY lane 0 B
A3	CSI0_DP0_A0	CSI0 D-PHY differential data 0 positive/CSI0 C-PHY lane 0 A
D4	CSI0_DN1_A1	CSI0 D-PHY differential data 1 negative/CSI0 C-PHY lane 1 A
C3	CSI0_DP1_C0	CSI0 D-PHY differential data 1 positive/CSI0 C-PHY lane 0 C
D8	CSI0_DN2_B2	CSI0 D-PHY differential data 2 negative/CSI0 C-PHY lane 2 B
F8	CSI0_DP2_A2	CSI0 D-PHY differential data 2 positive/CSI0 C-PHY lane 2 A
E7	CSI0_DN3_NC	CSI0 D-PHY differential data 3 negative/Not Connect
C7	CSI0_DP3_C2	CSI0 D-PHY differential data 3 positive/CSI0 C-PHY lane 2 C
H2	CSI1_CKN_C1	CSI1 D-PHY differential clock 0 negative/CSI1 C-PHY lane 1 C
G1	CSI1_CKP_B1	CSI1 D-PHY differential clock 0 positive/CSI1 C-PHY lane 1 B
F4	CSI1_DN0_B0	CSI1 D-PHY differential data 0 negative/CSI1 C-PHY lane 0 B
F6	CSI1_DP0_A0	CSI1 D-PHY differential data 0 positive/CSI1 C-PHY lane 0 A

Ball No	Ball Name	Description
G3	CSI1_DN1_A1	CSI1 D-PHY differential data 1 negative/CSI1 C-PHY lane 1 A
G5	CSI1_DP1_C0	CSI1 D-PHY differential data 1 positive/CSI1 C-PHY lane 0 C
E1	CSI1_DN2_B2	CSI1 D-PHY differential data 2 negative/CSI1 C-PHY lane 2 B
F2	CSI1_DP2_A2	CSI1 D-PHY differential data 2 positive/CSI1 C-PHY lane 2 A
C1	CSI1_DN3_NC	CSI1 D-PHY differential data 3 negative/Not Connect
D2	CSI1_DP3_C2	CSI1 D-PHY differential data 3 positive/CSI1 C-PHY lane 2 C
A7	CSI2_CKN0	CSI2 differential clock 0 negative
B6	CSI2_CKP0	CSI2 differential clock 0 positive
A5	CSI2_CKN1	CSI2 differential clock 1 negative
B4	CSI2_CKP1	CSI2 differential clock 1 positive
A11	CSI2_DN0	CSI2 differential data 0 negative
B10	CSI2_DP0	CSI2 differential data 0 positive
A9	CSI2_DN1	CSI2 differential data 1 negative
B8	CSI2_DP1	CSI2 differential data 1 positive
C9	CSI2_DN2	CSI2 differential data 2 negative
E9	CSI2_DP2	CSI2 differential data 2 positive
D10	CSI2_DN3	CSI2 differential data 3 negative
F10	CSI2_DP3	CSI2 differential data 3 positive
C13	CSI3_CKN0	CSI3 differential clock 0 negative
E13	CSI3_CKP0	CSI3 differential clock 0 positive
F14	CSI3_CKN1	CSI3 differential clock 1 negative
D14	CSI3_CKP1	CSI3 differential clock 1 positive
D12	CSI3_DN0	CSI3 differential data 0 negative
F12	CSI3_DP0	CSI3 differential data 0 positive
A13	CSI3_DN1	CSI3 differential data 1 negative
B12	CSI3_DP1	CSI3 differential data 1 positive
B14	CSI3_DN2	CSI3 differential data 2 negative
A15	CSI3_DP2	CSI3 differential data 2 positive
C15	CSI3_DN3	CSI3 differential data 3 negative
E15	CSI3_DP3	CSI3 differential data 3 positive
Clock Interface		

Ball No	Ball Name	Description
Y2	CLK26M_OUT_PMIC	26M clock output for PMIC
Y4	CLK52M_IN	52M clock input for AP
F36	CLK52M_IN_PCIE	52M clock input for PCIE
AT34	CLK52M_IN_USB	52M clock input for USB and UFS
DSI Interface		
AT28	DSI0_CKN	DSI0 differential clock negative
AV28	DSI0_CKP	DSI0 differential clock positive
AY28	DSI0_DN0	DSI0 differential data 0 negative
BA29	DSI0_DP0	DSI0 differential data 0 positive
AW29	DSI0_DN1	DSI0 differential data 1 negative
AU29	DSI0_DP1	DSI0 differential data 1 positive
AW27	DSI0_DN2	DSI0 differential data 2 negative
AU27	DSI0_DP2	DSI0 differential data 2 positive
AY26	DSI0_DN3	DSI0 differential data 3 negative
BA27	DSI0_DP3	DSI0 differential data 3 positive
AT26	DSI1_CKN	DSI1 differential clock negative
AP26	DSI1_CKP	DSI1 differential clock positive
AW23	DSI1_DN0	DSI1 differential data 0 negative
AU23	DSI1_DP0	DSI1 differential data 0 positive
AT24	DSI1_DN1	DSI1 differential data 1 negative
AV24	DSI1_DP1	DSI1 differential data 1 positive
AU25	DSI1_DN2	DSI1 differential data 2 negative
AW25	DSI1_DP2	DSI1 differential data 2 positive
BA25	DSI1_DN3	DSI1 differential data 3 negative
AY24	DSI1_DP3	DSI1 differential data 3 positive
LVDSRF Analog Interface		
AC5	LVDSRF_DL0_N	LVDSRF Downlink data 0 negative
AC3	LVDSRF_DL0_P	LVDSRF Downlink data 0 positive
AB6	LVDSRF_DL1_N	LVDSRF Downlink data 1 negative
AB4	LVDSRF_DL1_P	LVDSRF Downlink data 1 positive
AA1	LVDSRF_DL2_N	LVDSRF Downlink data 2 negative

Ball No	Ball Name	Description
AB2	LVDSRF_DL2_P	LVDSRF Downlink data 2 positive
AC1	LVDSRF_DL3_N	LVDSRF Downlink data 3 negative
AD2	LVDSRF_DL3_P	LVDSRF Downlink data 3 positive
AF6	LVDSRF_UL0_N	LVDSRF Uplink data 0 negative
AF4	LVDSRF_UL0_P	LVDSRF Uplink data 0 positive
AE1	LVDSRF_UL1_N	LVDSRF Uplink data 1 negative
AF2	LVDSRF_UL1_P	LVDSRF Uplink data 1 positive
AE3	LVDSRF_UL2_N	LVDSRF Uplink data 2 negative
AE5	LVDSRF_UL2_P	LVDSRF Uplink data 2 positive
AA7	LVDSRF_TEST_HS_ON	LVDSRF highspeed negative test ball
AB8	LVDSRF_TEST_HS_OP	LVDSRF highspeed positive test ball
AC7	LVDSRF_TEST_LS	LVDSRF lowspeed test ball
APC Interface		
W1	APCOUT0	APC output 0
W5	APCOUT1	APC output 1
USB2.0 Interface		
AR31	USB_DM	USB2.0 data negative
AP30	USB_DP	USB2.0 data positive
AY30	USB_REXT	USB2.0 calibration pin
USB3.1 Interface		
AT32	SSUSB_RXN1	USB3.1 RX 1 negative
AV32	SSUSB_RXP1	USB3.1 RX 1 positive
AY32	SSUSB_RXN2	USB3.1 RX 2 negative
BA33	SSUSB_RXP2	USB3.1 RX 2 positive
BA35	SSUSB_TXN1	USB3.1 TX 1 negative
AY36	SSUSB_TXP1	USB3.1 TX 1 positive
AW35	SSUSB_TXN2	USB3.1 TX 2 negative
AU35	SSUSB_TXP2	USB3.1 TX 2 positive
BA31	SSUSB_REXT	USB3.1 calibration pin
AW31	USB_DP_AUX1	USB DP AUX channel 1
AU31	USB_DP_AUX2	USB DP AUX channel 2

Ball No	Ball Name	Description
UFS Interface		
AR35	UFS_CLK	UFS reference clock output
AP34	UFS_RST	UFS reset output
AU37	UFS_RXN0	UFS RX0 negative
AW37	UFS_RXP0	UFS RX0 positive
AU39	UFS_RXN1	UFS RX1 negative
AW39	UFS_RXP1	UFS RX1 positive
BA41	UFS_TXN0	UFS TX0 negative
AY40	UFS_TXP0	UFS TX0 positive
BA39	UFS_TXN1	UFS TX1 negative
AY38	UFS_TXP1	UFS TX1 positive
AR37	UFS_REXT	UFS calibration pin
PCIe Interface		
D34	PCIE_CKN	PCIe differential clock negative
F34	PCIE_CKP	PCIe differential clock positive
C33	PCIE_RXN	PCIe RX negative
E33	PCIE_RXP	PCIe RX positive
D36	PCIE_TXN	PCIe TX negative
E35	PCIE_TXP	PCIe TX positive
K34	PCIE_REXT	PCIe calibration pin
System Power Input		
J9	AVDD1V2_CSI	CSI 1.2V analog power supply
J11	AVDD1V2_CSI	CSI 1.2V analog power supply
AM30	AVDD1V2_DSI	DSI 1.2V analog power supply
AC9	AVDD1V2_LVDSRF	LVDSRF 1.2V analog power supply
H32	AVDD1V2_PCIE	PCIe 1.2V analog power supply
AN35	AVDD1V2_UFS	UFS 1.2V analog power supply
AM32	AVDD1V2_USB	USB3.1 1.2V analog power supply
W9	AVDD1V8_BB	Baseband DAC & PLL 1.8V analog power supply
J13	AVDD1V8_CSI	CSI 1.8V analog power supply
W35	AVDD1V8_DPLL	DDR PLL 1.8V analog power supply

Ball No	Ball Name	Description
AP28	AVDD1V8_DSI	DSI 1.8V analog power supply
H36	AVDD1V8_EFU	EFUSE 1.8V analog power supply
AA9	AVDD1V8_LVDSRF	LVDSRF 1.8V analog power supply
H30	AVDD1V8_MPLLBIG	MPLLBIG 1.8V analog power supply
V14	AVDD1V8_MPLLLIT	MPLLLIT 1.8V analog power supply
J23	AVDD1V8_MPLLMID	MPLLMID 1.8V analog power supply
H34	AVDD1V8_PCIEPLL	PCIEPLL 1.8V analog power supply
AN29	AVDD1V8_USB	USB2.0 1.8V analog power supply
AM34	AVDD1V8_USBPLL	USB3.1 PLL 1.8V analog power supply
AN31	AVDD3V3_USB	USB2.0 3.3V analog power supply
J33	VDD0V75_PCIE	PCIe 0.75V digital power supply
AL35	VDD0V75_UFS	UFS 0.75V digital power supply
AL33	VDD0V75_USB	USB3.1 0.75V digital power supply
L33	VDD1V8_ISE	ISE 1.8V digital power supply
AK36	VEMMCIO	Power supply for eMMC IO
F30	VIO1V8	Power supply for digital IO
G25	VIO1V8	Power supply for digital IO
G31	VIO1V8	Power supply for digital IO
J37	VIO1V8	Power supply for digital IO
T8	VIO1V8	Power supply for digital IO
AF8	VIO1V8	Power supply for digital IO
AP18	VIO1V8	Power supply for digital IO
AT14	VSD0	Power supply for SDIO0 IO
AR13	VSD2	Power supply for SDIO2 IO
AR15	VSIM0	Power supply for SIM card 0
AP14	VSIM1	Power supply for SIM card 1
K8	VDDAI	Power supply for AI
K10	VDDAI	Power supply for AI
L9	VDDAI	Power supply for AI
M8	VDDAI	Power supply for AI
N9	VDDAI	Power supply for AI

Ball No	Ball Name	Description
R9	VDDAI	Power supply for AI
H24	VDDCORE	Power supply for Core domain
J19	VDDCORE	Power supply for Core domain
L35	VDDCORE	Power supply for Core domain
M34	VDDCORE	Power supply for Core domain
N33	VDDCORE	Power supply for Core domain
U9	VDDCORE	Power supply for Core domain
U11	VDDCORE	Power supply for Core domain
U19	VDDCORE	Power supply for Core domain
V30	VDDCORE	Power supply for Core domain
W17	VDDCORE	Power supply for Core domain
W19	VDDCORE	Power supply for Core domain
W21	VDDCORE	Power supply for Core domain
W23	VDDCORE	Power supply for Core domain
W25	VDDCORE	Power supply for Core domain
W27	VDDCORE	Power supply for Core domain
W29	VDDCORE	Power supply for Core domain
AA23	VDDCORE	Power supply for Core domain
AB22	VDDCORE	Power supply for Core domain
AC21	VDDCORE	Power supply for Core domain
AD8	VDDCORE	Power supply for Core domain
AD22	VDDCORE	Power supply for Core domain
AE23	VDDCORE	Power supply for Core domain
AG23	VDDCORE	Power supply for Core domain
AH24	VDDCORE	Power supply for Core domain
AH30	VDDCORE	Power supply for Core domain
AH32	VDDCORE	Power supply for Core domain
AJ33	VDDCORE	Power supply for Core domain
AN17	VDDCORE	Power supply for Core domain
AN27	VDDCORE	Power supply for Core domain
J29	VDDCPU_BIG	Power supply for Big core CPU

Ball No	Ball Name	Description
K30	VDDCPU_BIG	Power supply for Big core CPU
L29	VDDCPU_BIG	Power supply for Big core CPU
M30	VDDCPU_BIG	Power supply for Big core CPU
N29	VDDCPU_BIG	Power supply for Big core CPU
P20	VDDCPU_LIT	Power supply for Little core CPU
P28	VDDCPU_LIT	Power supply for Little core CPU
R21	VDDCPU_LIT	Power supply for Little core CPU
R25	VDDCPU_LIT	Power supply for Little core CPU
R29	VDDCPU_LIT	Power supply for Little core CPU
T24	VDDCPU_LIT	Power supply for Little core CPU
U21	VDDCPU_LIT	Power supply for Little core CPU
U25	VDDCPU_LIT	Power supply for Little core CPU
U29	VDDCPU_LIT	Power supply for Little core CPU
J21	VDDCPU_MID	Power supply for Middle core CPU
J25	VDDCPU_MID	Power supply for Middle core CPU
K22	VDDCPU_MID	Power supply for Middle core CPU
K26	VDDCPU_MID	Power supply for Middle core CPU
L21	VDDCPU_MID	Power supply for Middle core CPU
L25	VDDCPU_MID	Power supply for Middle core CPU
M22	VDDCPU_MID	Power supply for Middle core CPU
M26	VDDCPU_MID	Power supply for Middle core CPU
Y32	VDDGPU	Power supply for GPU
AA27	VDDGPU	Power supply for GPU
AA31	VDDGPU	Power supply for GPU
AB26	VDDGPU	Power supply for GPU
AB32	VDDGPU	Power supply for GPU
AC25	VDDGPU	Power supply for GPU
AC33	VDDGPU	Power supply for GPU
AD26	VDDGPU	Power supply for GPU
AD32	VDDGPU	Power supply for GPU
AE27	VDDGPU	Power supply for GPU

Ball No	Ball Name	Description
AE31	VDDGPU	Power supply for GPU
AF26	VDDGPU	Power supply for GPU
AF28	VDDGPU	Power supply for GPU
AF32	VDDGPU	Power supply for GPU
AG25	VDDGPU	Power supply for GPU
AG33	VDDGPU	Power supply for GPU
T36	VDDMEM	Power supply for LPDDR4X core
AD36	VDDMEM	Power supply for LPDDR4X core
M36	VDDMEMQ	Power supply for LPDDR4X IO
N35	VDDMEMQ	Power supply for LPDDR4X IO
P36	VDDMEMQ	Power supply for LPDDR4X IO
R35	VDDMEMQ	Power supply for LPDDR4X IO
U35	VDDMEMQ	Power supply for LPDDR4X IO
Y36	VDDMEMQ	Power supply for LPDDR4X IO
AA35	VDDMEMQ	Power supply for LPDDR4X IO
AB36	VDDMEMQ	Power supply for LPDDR4X IO
AC35	VDDMEMQ	Power supply for LPDDR4X IO
AE35	VDDMEMQ	Power supply for LPDDR4X IO
K12	VDDMM	Power supply for Multi-Media
K18	VDDMM	Power supply for Multi-Media
L13	VDDMM	Power supply for Multi-Media
L17	VDDMM	Power supply for Multi-Media
M14	VDDMM	Power supply for Multi-Media
M16	VDDMM	Power supply for Multi-Media
N13	VDDMM	Power supply for Multi-Media
N17	VDDMM	Power supply for Multi-Media
P14	VDDMM	Power supply for Multi-Media
P18	VDDMM	Power supply for Multi-Media
R17	VDDMM	Power supply for Multi-Media
T16	VDDMM	Power supply for Multi-Media
U15	VDDMM	Power supply for Multi-Media

Ball No	Ball Name	Description
U17	VDDMM	Power supply for Multi-Media
AJ23	VDDMM	Power supply for Multi-Media
AJ25	VDDMM	Power supply for Multi-Media
AJ29	VDDMM	Power supply for Multi-Media
AK24	VDDMM	Power supply for Multi-Media
AK30	VDDMM	Power supply for Multi-Media
AL23	VDDMM	Power supply for Multi-Media
AL29	VDDMM	Power supply for Multi-Media
AM22	VDDMM	Power supply for Multi-Media
AM24	VDDMM	Power supply for Multi-Media
W11	VDDMODEM	Power supply for Modem
Y12	VDDMODEM	Power supply for Modem
AA11	VDDMODEM	Power supply for Modem
AA17	VDDMODEM	Power supply for Modem
AB16	VDDMODEM	Power supply for Modem
AC11	VDDMODEM	Power supply for Modem
AC17	VDDMODEM	Power supply for Modem
AD10	VDDMODEM	Power supply for Modem
AD16	VDDMODEM	Power supply for Modem
AE11	VDDMODEM	Power supply for Modem
AE15	VDDMODEM	Power supply for Modem
AE21	VDDMODEM	Power supply for Modem
AF10	VDDMODEM	Power supply for Modem
AF14	VDDMODEM	Power supply for Modem
AF20	VDDMODEM	Power supply for Modem
AG19	VDDMODEM	Power supply for Modem
AH10	VDDMODEM	Power supply for Modem
AH14	VDDMODEM	Power supply for Modem
AH20	VDDMODEM	Power supply for Modem
AJ11	VDDMODEM	Power supply for Modem
AJ15	VDDMODEM	Power supply for Modem

Ball No	Ball Name	Description
AJ19	VDDMODEM	Power supply for Modem
AK10	VDDMODEM	Power supply for Modem
AK14	VDDMODEM	Power supply for Modem
AK20	VDDMODEM	Power supply for Modem
AL9	VDDMODEM	Power supply for Modem
AL15	VDDMODEM	Power supply for Modem
AL19	VDDMODEM	Power supply for Modem
AM10	VDDMODEM	Power supply for Modem
AM14	VDDMODEM	Power supply for Modem
AN9	VDDMODEM	Power supply for Modem
AN15	VDDMODEM	Power supply for Modem
P34	VDDPUB	Power supply for DDR controller
T34	VDDPUB	Power supply for DDR controller
V34	VDDPUB	Power supply for DDR controller
Y34	VDDPUB	Power supply for DDR controller
AB34	VDDPUB	Power supply for DDR controller
AD34	VDDPUB	Power supply for DDR controller
AF34	VDDPUB	Power supply for DDR controller
J17	VDDSRAM	Power supply for Dual-rail SRAM
K16	VDDSRAM	Power supply for Dual-rail SRAM
M12	VDDSRAM	Power supply for Dual-rail SRAM
T14	VDDSRAM	Power supply for Dual-rail SRAM
W31	VDDSRAM	Power supply for Dual-rail SRAM
Y14	VDDSRAM	Power supply for Dual-rail SRAM
Y18	VDDSRAM	Power supply for Dual-rail SRAM
Y30	VDDSRAM	Power supply for Dual-rail SRAM
AA13	VDDSRAM	Power supply for Dual-rail SRAM
AA19	VDDSRAM	Power supply for Dual-rail SRAM
AB14	VDDSRAM	Power supply for Dual-rail SRAM
AB18	VDDSRAM	Power supply for Dual-rail SRAM
AB30	VDDSRAM	Power supply for Dual-rail SRAM

Ball No	Ball Name	Description
AC29	VDDSRAM	Power supply for Dual-rail SRAM
AD14	VDDSRAM	Power supply for Dual-rail SRAM
AD18	VDDSRAM	Power supply for Dual-rail SRAM
AE13	VDDSRAM	Power supply for Dual-rail SRAM
AE19	VDDSRAM	Power supply for Dual-rail SRAM
AE29	VDDSRAM	Power supply for Dual-rail SRAM
AF18	VDDSRAM	Power supply for Dual-rail SRAM
AG13	VDDSRAM	Power supply for Dual-rail SRAM
AG17	VDDSRAM	Power supply for Dual-rail SRAM
AH18	VDDSRAM	Power supply for Dual-rail SRAM
AJ13	VDDSRAM	Power supply for Dual-rail SRAM
AK18	VDDSRAM	Power supply for Dual-rail SRAM
AK26	VDDSRAM	Power supply for Dual-rail SRAM
AL13	VDDSRAM	Power supply for Dual-rail SRAM
AL27	VDDSRAM	Power supply for Dual-rail SRAM
AM12	VDDSRAM	Power supply for Dual-rail SRAM
AM18	VDDSRAM	Power supply for Dual-rail SRAM
AM26	VDDSRAM	Power supply for Dual-rail SRAM
P26	VDDSRAM_CPU_LIT	Power supply for Dual-rail SRAM of Little core CPU
T26	VDDSRAM_CPU_LIT	Power supply for Dual-rail SRAM of Little core CPU
L23	VDDSRAM_CPU_MID	Power supply for Dual-rail SRAM of Middle core CPU
Ground		
W3	AVSS_BB	Analog ground of Baseband DAC and PLL
B16	AVSS_CSI	Analog ground of CSI
C11	AVSS_CSI	Analog ground of CSI
D6	AVSS_CSI	Analog ground of CSI
D16	AVSS_CSI	Analog ground of CSI
E3	AVSS_CSI	Analog ground of CSI
E11	AVSS_CSI	Analog ground of CSI
G7	AVSS_CSI	Analog ground of CSI
G9	AVSS_CSI	Analog ground of CSI

Ball No	Ball Name	Description
G11	AVSS_CSI	Analog ground of CSI
G13	AVSS_CSI	Analog ground of CSI
H4	AVSS_CSI	Analog ground of CSI
J1	AVSS_CSI	Analog ground of CSI
G33	AVSS_DDC	Analog ground of DDC
V36	AVSS_DPLL	Analog ground of DDR PLL
AR25	AVSS_DSI	Analog ground of DSI
AR27	AVSS_DSI	Analog ground of DSI
AR29	AVSS_DSI	Analog ground of DSI
AV26	AVSS_DSI	Analog ground of DSI
BA23	AVSS_DSI	Analog ground of DSI
R33	AVSS_ISE	Analog ground of ISE
AA3	AVSS_LVDSRF	Analog ground of LVDSRF
AA5	AVSS_LVDSRF	Analog ground of LVDSRF
AD4	AVSS_LVDSRF	Analog ground of LVDSRF
AD6	AVSS_LVDSRF	Analog ground of LVDSRF
AE7	AVSS_LVDSRF	Analog ground of LVDSRF
AG1	AVSS_LVDSRF	Analog ground of LVDSRF
AG5	AVSS_LVDSRF	Analog ground of LVDSRF
H28	AVSS_MPLLBIG	Analog ground of MPLLBIG
Y20	AVSS_MPLLLIT	Analog ground of MPLLLIT
H22	AVSS_MPLLMID	Analog ground of MPLLMID
E37	AVSS_PCIEPLL	Analog ground of PCIEPLL
G35	AVSS_PCIEPLL	Analog ground of PCIEPLL
J35	AVSS_RCO	Analog ground of RCO
T10	AVSS_THM	Analog ground of Thermal sensor
AR33	AVSS_USBPLL	Analog ground of USBPLL
AT36	AVSS_USBPLL	Analog ground of USBPLL
C35	VSS	Ground
D24	VSS	Ground
D32	VSS	Ground

Ball No	Ball Name	Description
D40	VSS	Ground
D42	VSS	Ground
E39	VSS	Ground
G37	VSS	Ground
J15	VSS	Ground
J27	VSS	Ground
J31	VSS	Ground
J39	VSS	Ground
J41	VSS	Ground
J43	VSS	Ground
K14	VSS	Ground
K20	VSS	Ground
K24	VSS	Ground
K28	VSS	Ground
K32	VSS	Ground
K36	VSS	Ground
L7	VSS	Ground
L11	VSS	Ground
L15	VSS	Ground
L19	VSS	Ground
L27	VSS	Ground
L31	VSS	Ground
L37	VSS	Ground
M10	VSS	Ground
M18	VSS	Ground
M20	VSS	Ground
M28	VSS	Ground
M32	VSS	Ground
N7	VSS	Ground
N15	VSS	Ground
N19	VSS	Ground

Ball No	Ball Name	Description
N21	VSS	Ground
N27	VSS	Ground
P8	VSS	Ground
P16	VSS	Ground
P38	VSS	Ground
P40	VSS	Ground
P42	VSS	Ground
R15	VSS	Ground
R19	VSS	Ground
R27	VSS	Ground
R37	VSS	Ground
R39	VSS	Ground
T18	VSS	Ground
T20	VSS	Ground
T22	VSS	Ground
T28	VSS	Ground
U13	VSS	Ground
U23	VSS	Ground
U27	VSS	Ground
U31	VSS	Ground
U33	VSS	Ground
U37	VSS	Ground
V12	VSS	Ground
V16	VSS	Ground
V18	VSS	Ground
V20	VSS	Ground
V22	VSS	Ground
V24	VSS	Ground
V26	VSS	Ground
V28	VSS	Ground
V32	VSS	Ground

Ball No	Ball Name	Description
W13	VSS	Ground
W15	VSS	Ground
W37	VSS	Ground
W39	VSS	Ground
W41	VSS	Ground
Y10	VSS	Ground
Y16	VSS	Ground
Y24	VSS	Ground
Y26	VSS	Ground
Y28	VSS	Ground
AA15	VSS	Ground
AA21	VSS	Ground
AA25	VSS	Ground
AA29	VSS	Ground
AA33	VSS	Ground
AA37	VSS	Ground
AB10	VSS	Ground
AB12	VSS	Ground
AB20	VSS	Ground
AB24	VSS	Ground
AB28	VSS	Ground
AC13	VSS	Ground
AC15	VSS	Ground
AC19	VSS	Ground
AC23	VSS	Ground
AC27	VSS	Ground
AC31	VSS	Ground
AC37	VSS	Ground
AC39	VSS	Ground
AD12	VSS	Ground
AD20	VSS	Ground

Ball No	Ball Name	Description
AD24	VSS	Ground
AD28	VSS	Ground
AD30	VSS	Ground
AD38	VSS	Ground
AD40	VSS	Ground
AD42	VSS	Ground
AE9	VSS	Ground
AE17	VSS	Ground
AE25	VSS	Ground
AE33	VSS	Ground
AE37	VSS	Ground
AF12	VSS	Ground
AF16	VSS	Ground
AF22	VSS	Ground
AF24	VSS	Ground
AF30	VSS	Ground
AF36	VSS	Ground
AG9	VSS	Ground
AG11	VSS	Ground
AG15	VSS	Ground
AG21	VSS	Ground
AG27	VSS	Ground
AG29	VSS	Ground
AG31	VSS	Ground
AG35	VSS	Ground
AG37	VSS	Ground
AH12	VSS	Ground
AH16	VSS	Ground
AH22	VSS	Ground
AH26	VSS	Ground
AH28	VSS	Ground

Ball No	Ball Name	Description
AH34	VSS	Ground
AH36	VSS	Ground
AJ9	VSS	Ground
AJ17	VSS	Ground
AJ21	VSS	Ground
AJ27	VSS	Ground
AJ31	VSS	Ground
AJ35	VSS	Ground
AJ37	VSS	Ground
AJ41	VSS	Ground
AJ43	VSS	Ground
AK12	VSS	Ground
AK16	VSS	Ground
AK22	VSS	Ground
AK28	VSS	Ground
AK32	VSS	Ground
AK34	VSS	Ground
AL11	VSS	Ground
AL17	VSS	Ground
AL21	VSS	Ground
AL25	VSS	Ground
AL31	VSS	Ground
AL37	VSS	Ground
AM16	VSS	Ground
AM20	VSS	Ground
AM28	VSS	Ground
AM36	VSS	Ground
AN11	VSS	Ground
AN13	VSS	Ground
AN19	VSS	Ground
AN21	VSS	Ground

Ball No	Ball Name	Description
AN25	VSS	Ground
AN37	VSS	Ground
AN39	VSS	Ground
AP32	VSS	Ground
AP36	VSS	Ground
AP40	VSS	Ground
AP42	VSS	Ground
AT30	VSS	Ground
AV6	VSS	Ground
AV30	VSS	Ground
AV34	VSS	Ground
AV36	VSS	Ground
AV40	VSS	Ground
AY34	VSS	Ground
BA37	VSS	Ground
AN23	VSS	Ground
NC		
A1	NC	Not Connect
A43	NC	Not Connect
BA1	NC	Not Connect
BA43	NC	Not Connect

3.5 Pin Multiplexed Function List

Table 3-3 Pin Multiplexed Function List

Ball No	Cell Name	Ball name	Function0	Type	Function1	Type	Function2	Type	Function3	Type
BA13	PDWUWS WCDGSDE	SIMCLK0	SIMCLK	O					GPIO1	I/O/T
BA11	PDWUWS WCDGSDE	SIMDATA0	SIMDATA	I/O/T					GPIO2	I/O/T

Ball No	Cell Name	Ball name	Function0	Type	Function1	Type	Function2	Type	Function3	Type
AY10	PDWUWS WCDGSDE	SIMRST0	SIMRST0	O					GPIO3	I/O/T
AP10	PDWUWS WCDGSDE	SIMCLK1	SIMCLK1	O					GPIO4	I/O/T
AT10	PDWUWS WCDGSDE	SIMDATA1	SIMDATA1	I/O/T					GPIO5	I/O/T
AR11	PDWUWS WCDGSDE	SIMRST1	SIMRST1	O					GPIO6	I/O/T
F26	SPSCBC2_8X_W	KEYIN0	KEYIN0	I/O/T	EXTINT2	I/O/T	SE_GPIO4	I/O/T	GPIO7	I/O/T
B26	SPSCBC2_8X_W	KEYIN1	KEYIN1	I/O/T	EXTINT3	I/O/T	SE_GPIO5	I/O/T	GPIO8	I/O/T
E25	SPSCBC2_8X_W	KEYIN2	KEYIN2	I/O/T	EXTINT4	I/O/T			GPIO9	I/O/T
AP22	SPSCBC2_8X	PWMA	PWMA	O					GPIO10	I/O/T
AV22	SPSCBC2_8X_W	LCM0_RSTN	LCM0_RSTN	O					GPIO11	I/O/T
AW21	SPSCBC2_8X_W	DSI0_TE	DSI0_TE	I					GPIO12	I/O/T
AV20	SPSCBC2_8X_W	EXTINT0	EXTINT0	I	KEYOUT5	I/O/T	SE_GPIO14	I/O/T	GPIO13	I/O/T
AU21	SPSCBC2_8X_W	EXTINT1	EXTINT1	I	KEYOUT6	I/O/T	SE_GPIO15	I/O/T	GPIO14	I/O/T
AR23	SPSCBC2_8X_W	SDA3	SDA3	I/O/T					GPIO15	I/O/T
AP24	SPSCBC2_8X_W	SCL3	SCL3	I/O/T					GPIO16	I/O/T
AV8	PDWUWS WCDGSDE	SD0_D3	SD0_D3	I/O/T					GPIO17	I/O/T
AU9	PDWUWS WCDGSDE	SD0_D2	SD0_D2	I/O/T					GPIO18	I/O/T
AW9	PDWUWS WCDGSDE	SD0_CMD	SD0_CMD	I/O/T					GPIO19	I/O/T
BA9	PDWUWS WCDGSDE	SD0_D0	SD0_D0	I/O/T					GPIO20	I/O/T
BA7	PDWUWS WCDGSDE	SD0_D1	SD0_D1	I/O/T					GPIO21	I/O/T

Ball No	Cell Name	Ball name	Function0	Type	Function1	Type	Function2	Type	Function3	Type
AY8	PDWUWS WCDGSDE	SD0_CLK	SD0_CLK	O/T					GPIO2 2	I/O/T
AT40	SPSEBC2_24X	EMMC_D0	EMMC_D0	I/O/T					GPIO2 3	I/O/T
AR41	SPSEBC2_24X	EMMC_CMD	EMMC_CMD	I/O/T					GPIO2 4	I/O/T
AY42	SPSEBC2_24X	EMMC_D6	EMMC_D6	I/O/T					GPIO2 5	I/O/T
AW41	SPSEBC2_24X	EMMC_D7	EMMC_D7	I/O/T					GPIO2 6	I/O/T
AU43	SPSEBC2_24X	EMMC_CLK	EMMC_CLK	O/T					GPIO2 7	I/O/T
AR39	SPSEBC2_24X	EMMC_D5	EMMC_D5	I/O/T					GPIO2 8	I/O/T
AV42	SPSEBC2_24X	EMMC_D4	EMMC_D4	I/O/T					GPIO2 9	I/O/T
AT42	SPSEBC2_24X	EMMC_DS	EMMC_DS	I					GPIO3 0	I/O/T
AP38	SPSEBC2_24X	EMMC_D3	EMMC_D3	I/O/T					GPIO3 1	I/O/T
AR43	SPSEBC2_24X	EMMC_RST	EMMC_RST	O					GPIO3 2	I/O/T
AW43	SPSEBC2_24X	EMMC_D1	EMMC_D1	I/O/T					GPIO3 3	I/O/T
AU41	SPSEBC2_24X	EMMC_D2	EMMC_D2	I/O/T					GPIO3 4	I/O/T
A17	SPSCBC2_8X	CMMC_LK0	CMMC_LK0	O			DBG_B US0(G1)	I/O/T	GPIO3 5	I/O/T
C17	SPSCBC2_8X	CMMC_LK1	CMMC_LK1	O			DBG_B US1(G1)	I/O/T	GPIO3 6	I/O/T
B18	SPSCBC2_8X	CMMC_LK2	CMMC_LK2	O	EXTINT 7(G1)	I/O/T	DBG_B US2(G1)	I/O/T	GPIO3 7	I/O/T
E17	SPSCBC2_8X	CMMC_LK3	CMMC_LK3	O	CLK_A UX3	O	DBG_B US3(G1)	I/O/T	GPIO3 8	I/O/T
J3	SPSCBC2_8X	CMRST0	CMRST0	I/O/T			DBG_B US4(G1)	I/O/T	GPIO3 9	I/O/T
K6	SPSCBC2_8X	CMRST1	CMRST1	I/O/T			DBG_B US5(G1)	I/O/T	GPIO4 0	I/O/T

Ball No	Cell Name	Ball name	Function0	Type	Function1	Type	Function2	Type	Function3	Type
J7	SPSCBC2_8X	CMRST2	CMRST2	I/O/T			DBG_BUS6(G1)	I/O/T	GPIO41	I/O/T
L1	SPSCBC2_8X	CMRST3	CMRST3	I/O/T			DBG_BUS7(G1)	I/O/T	GPIO42	I/O/T
J5	SPSCBC2_8X	CMRST4	CMRST4	I/O/T	KEYIN3	I/O/T	DBG_BUS8(G1)	I/O/T	GPIO43	I/O/T
K2	SPSCBC2_8X	CMRST5	CMRST5	I/O/T	KEYIN4	I/O/T	DBG_BUS9(G1)	I/O/T	GPIO44	I/O/T
G17	SPSCBC2_8X	CMPD0	CMPD0	I/O/T			DBG_BUS10(G1)	I/O/T	GPIO45	I/O/T
K4	SPSCBC2_8X	CMPD1	CMPD1	I/O/T			DBG_BUS11(G1)	I/O/T	GPIO46	I/O/T
D18	SPSCBC2_8X	CMPD2	CMPD2	I/O/T			DBG_BUS12(G1)	I/O/T	GPIO47	I/O/T
H16	SPSCBC2_8X	CMPD3	CMPD3	I/O/T			DBG_BUS13(G1)	I/O/T	GPIO48	I/O/T
F18	SPSCBC2_8X	CMPD4	CMPD4	I/O/T	KEYIN5	I/O/T	DBG_BUS14(G1)	I/O/T	GPIO49	I/O/T
H18	SPSCBC2_8X	CMPD5	CMPD5	I/O/T	KEYIN6	I/O/T	DBG_BUS15(G1)	I/O/T	GPIO50	I/O/T
A19	SPSCBC2_8X	SCL0	SCL0	I/O/T			DBG_BUS16(G1)	I/O/T	GPIO51	I/O/T
B20	SPSCBC2_8X	SDA0	SDA0	I/O/T			DBG_BUS17(G1)	I/O/T	GPIO52	I/O/T
C19	SPSCBC2_8X	SCL1	SCL1	I/O/T			DBG_BUS18(G1)	I/O/T	GPIO53	I/O/T
E19	SPSCBC2_8X	SDA1	SDA1	I/O/T			DBG_BUS19(G1)	I/O/T	GPIO54	I/O/T
C21	SPSCBC2_8X	SCL8	SCL8	I/O/T	KEYOUT3	I/O/T	DBG_BUS20(G1)	I/O/T	GPIO55	I/O/T

Ball No	Cell Name	Ball name	Function0	Type	Function1	Type	Function2	Type	Function3	Type
D20	SPSCBC2_8X	SDA8	SDA8	I/O/T	KEYOUT4	I/O/T	DBG_BUS21(G1)	I/O/T	GPIO56	I/O/T
H20	SPSCBC2_8X	SCL9	SCL9	I/O/T	EXTINT14(G0)	I/O/T	DBG_BUS22(G1)	I/O/T	GPIO57	I/O/T
G19	SPSCBC2_8X	SDA9	SDA9	I/O/T	EXTINT15(G0)	I/O/T	DBG_BUS23(G1)	I/O/T	GPIO58	I/O/T
A21	SPSCBC2_8X	SPI3_CS_N	SPI3_CS_N	I/O/T			ESE_GPIO6	I/O/T	GPIO59	I/O/T
B22	SPSCBC2_8X	SPI3_CLK	SPI3_CLK	I/O/T			ESE_GPIO7	I/O/T	GPIO60	I/O/T
E21	SPSCBC2_8X	SPI3_DI	SPI3_DI	I/O/T			CM4_GPIO2	I/O/T	GPIO61	I/O/T
D22	SPSCBC2_8X	SPI3_DO	SPI3_DO	I/O/T			ESE_GPIO5	I/O/T	GPIO62	I/O/T
AJ5	SPSCBC2_8X_SR	RFSPISDA0	RFSPISDA0	I/O/T					GPIO63	I/O/T
AH4	SPSCBC2_8X_SR	RFSPISDA1	RFSPISDA1	I/O/T					GPIO64	I/O/T
AJ3	SPSCBC2_8X_SR	RFSPISDA2	RFSPISDA2	I/O/T					GPIO65	I/O/T
AK2	SPSCBC2_8X_SR	RFSPISDA3	RFSPISDA3	I/O/T					GPIO66	I/O/T
AJ1	SPSCBC2_8X_SR	RFSPISCK	RFSPISCK	O/T					GPIO67	I/O/T
AH2	SPSCBC2_8X_SR	RFSPISEN	RFSPISEN	I/O/T					GPIO68	I/O/T
AP6	SPSCBC2_8X	RFCTL0(G0)	RFCTL0(G0)	I/O/T			DBG_BUS0(G0)	I/O/T	GPIO69	I/O/T
AP8	SPSCBC2_8X	RFCTL1(G0)	RFCTL1(G0)	I/O/T			DBG_BUS1(G0)	I/O/T	GPIO70	I/O/T
AR7	SPSCBC2_8X	RFCTL2(G0)	RFCTL2(G0)	I/O/T			DBG_BUS2(G0)	I/O/T	GPIO71	I/O/T
AU7	SPSCBC2_8X	RFCTL3(G0)	RFCTL3(G0)	I/O/T			DBG_BUS3(G0)	I/O/T	GPIO72	I/O/T
AR5	SPSCBC2_8X	RFCTL4(G0)	RFCTL4(G0)	I/O/T			DBG_BUS4(G0)	I/O/T	GPIO73	I/O/T

Ball No	Cell Name	Ball name	Function0	Type	Function1	Type	Function2	Type	Function3	Type
AT6	SPSCBC2_8X	RFCTL5 (G0)	RFCTL5 (G0)	I/O/T			DBG_B US5(G0)	I/O/T	GPIO7 4	I/O/T
AU5	SPSCBC2_8X	RFCTL6 (G0)	RFCTL6 (G0)	I/O/T			DBG_B US6(G0)	I/O/T	GPIO7 5	I/O/T
AT4	SPSCBC2_8X	RFCTL7 (G0)	RFCTL7 (G0)	I/O/T			DBG_B US7(G0)	I/O/T	GPIO7 6	I/O/T
AU3	SPSCBC2_8X	RFCTL8 (G0)	RFCTL8 (G0)	I/O/T			DBG_B US8(G0)	I/O/T	GPIO7 7	I/O/T
AV4	SPSCBC2_8X	RFCTL9 (G0)	RFCTL9 (G0)	I/O/T			DBG_B US9(G0)	I/O/T	GPIO7 8	I/O/T
AY4	SPSCBC2_8X	RFCTL10(G0)	RFCTL10(G0)	I/O/T			DBG_B US10(G0)	I/O/T	GPIO7 9	I/O/T
AY2	SPSCBC2_8X	RFCTL11(G0)	RFCTL11(G0)	I/O/T			DBG_B US11(G0)	I/O/T	GPIO8 0	I/O/T
AW3	SPSCBC2_8X	RFCTL12(G0)	RFCTL12(G0)	I/O/T			DBG_B US12(G0)	I/O/T	GPIO8 1	I/O/T
AW1	SPSCBC2_8X	RFCTL13(G0)	RFCTL13(G0)	I/O/T	WIFI_COEXIST	I/O/T	DBG_B US13(G0)	I/O/T	GPIO8 2	I/O/T
AV2	SPSCBC2_8X	RFCTL14(G0)	RFCTL14(G0)	I/O/T	UART_matrix.INF3.TXD	O	DBG_B US14(G0)	I/O/T	GPIO8 3	I/O/T
AU1	SPSCBC2_8X	RFCTL15(G0)	RFCTL15(G0)	I/O/T	UART_matrix.INF3.RXD	I	DBG_B US15(G0)	I/O/T	GPIO8 4	I/O/T
AL5	SPSCBC2_8X_SR	RFFE0_SCK	RFFE0_SCK	I/O/T	RFCTL0 (G1)	I/O/T	DBG_B US16(G0)	I/O/T	GPIO8 5	I/O/T
AM4	SPSCBC2_8X_SR	RFFE0_SDA	RFFE0_SDA	I/O/T	RFCTL1 (G1)	I/O/T	DBG_B US17(G0)	I/O/T	GPIO8 6	I/O/T
AK6	SPSCBC2_8X_SR	RFFE1_SCK	RFFE1_SCK	I/O/T	RFCTL2 (G1)	I/O/T	DBG_B US18(G0)	I/O/T	GPIO8 7	I/O/T
AK8	SPSCBC2_8X_SR	RFFE1_SDA	RFFE1_SDA	I/O/T	RFCTL3 (G1)	I/O/T	DBG_B US19(G0)	I/O/T	GPIO8 8	I/O/T

Ball No	Cell Name	Ball name	Function0	Type	Function1	Type	Function2	Type	Function3	Type
AK4	SPSCBC2_8X_SR	RFFE2_SCK	RFFE2_SCK	I/O/T	RFCTL4(G1)	I/O/T	DBG_BUS20(G0)	I/O/T	GPIO89	I/O/T
AL3	SPSCBC2_8X_SR	RFFE2_SDA	RFFE2_SDA	I/O/T	RFCTL5(G1)	I/O/T	DBG_BUS21(G0)	I/O/T	GPIO90	I/O/T
AN5	SPSCBC2_8X_SR	RFFE3_SCK	RFFE3_SCK	I/O/T	RFCTL6(G1)	I/O/T	DBG_BUS22(G0)	I/O/T	GPIO91	I/O/T
AN3	SPSCBC2_8X_SR	RFFE3_SDA	RFFE3_SDA	I/O/T	RFCTL7(G1)	I/O/T	DBG_BUS23(G0)	I/O/T	GPIO92	I/O/T
AP4	SPSCBC2_8X_SR	RFFE4_SCK	RFFE4_SCK	I/O/T	RFCTL8(G1)	I/O/T	DBG_BUS24(G0)	I/O/T	GPIO93	I/O/T
AR3	SPSCBC2_8X_SR	RFFE4_SDA	RFFE4_SDA	I/O/T	RFCTL9(G1)	I/O/T	DBG_BUS25(G0)	I/O/T	GPIO94	I/O/T
AT2	SPSCBC2_8X_SR	RFFE5_SCK	RFFE5_SCK	I/O/T	RFCTL10(G1)	I/O/T	DBG_BUS26(G0)	I/O/T	GPIO95	I/O/T
AR1	SPSCBC2_8X_SR	RFFE5_SDA	RFFE5_SDA	I/O/T	RFCTL11(G1)	I/O/T	DBG_BUS27(G0)	I/O/T	GPIO96	I/O/T
AN1	SPSCBC2_8X_SR	RFFE6_SCK	RFFE6_SCK	I/O/T	RFCTL12(G1)	I/O/T	DBG_BUS28(G0)	I/O/T	GPIO97	I/O/T
AP2	SPSCBC2_8X_SR	RFFE6_SDA	RFFE6_SDA	I/O/T	RFCTL13(G1)	I/O/T	DBG_BUS29(G0)	I/O/T	GPIO98	I/O/T
AM2	SPSCBC2_8X_SR	RFFE7_SCK	RFFE7_SCK	I/O/T	RFCTL14(G1)	I/O/T	DBG_BUS30(G0)	I/O/T	GPIO99	I/O/T
AL1	SPSCBC2_8X_SR	RFFE7_SDA	RFFE7_SDA	I/O/T	RFCTL15(G1)	I/O/T	DBG_BUS31(G0)	I/O/T	GPIO100	I/O/T
AH6	SPSCBC2_8X	LVDSRF_ADDAC0_ON	LVDSRF_ADDAC0_ON	I/O/T					GPIO101	I/O/T

Ball No	Cell Name	Ball name	Function0	Type	Function1	Type	Function2	Type	Function3	Type
AJ7	SPSCBC2_8X	LVDSR_F_ADDAC1_ON	LVDSR_F_ADDAC1_ON	I/O/T					GPIO102	I/O/T
AH8	SPSCBC2_8X	LVDSR_F_ADDAC2_ON	LVDSR_F_ADDAC2_ON	I/O/T					GPIO103	I/O/T
AG7	SPSCBC2_8X	LVDSR_F_ADDAC3_ON	LVDSR_F_ADDAC3_ON	I/O/T					GPIO104	I/O/T
A23	SPSCBC2_8X	SPI0_CS_N	SPI0_CS_N	I/O/T			EXTINT5(G0)	I/O/T	GPIO105	I/O/T
B24	SPSCBC2_8X	SPI0_DO	SPI0_DO	I/O/T			EXTINT6(G0)	I/O/T	GPIO106	I/O/T
C23	SPSCBC2_8X	SPI0_DI	SPI0_DI	I/O/T			EXTINT7(G0)	I/O/T	GPIO107	I/O/T
A25	SPSCBC2_8X	SPI0_CLK	SPI0_CLK	I/O/T			EXTINT8(G0)	I/O/T	GPIO108	I/O/T
AV12	SPSCBC2_8X	IIS1DI	IIS1DI	I	SE_GPIO0	I/O/T	EXTINT13	I/O/T	GPIO109	I/O/T
AW11	SPSCBC2_8X	IIS1DO	IIS1DO	O/T	SE_GPIO1	I/O/T	CM4_GPIO3(G0)	I/O/T	GPIO110	I/O/T
AV10	SPSCBC2_8X	IIS1CLK	IIS1CLK	I/O/T	SE_GPIO2	I/O/T	CM4_GPIO4(G0)	I/O/T	GPIO111	I/O/T
AU11	SPSCBC2_8X	IIS1LRC_K	IIS1LRC_K	I/O/T	SE_GPIO3	I/O/T			GPIO112	I/O/T
AT12	SPSCBC2_8X	SCL4	SCL4	I/O/T					GPIO113	I/O/T
AP12	SPSCBC2_8X	SDA4	SDA4	I/O/T					GPIO114	I/O/T
AY14	SPSCBC2_8X	CLK_AUX1	CLK_AUX1	O					GPIO115	I/O/T
BA5	PDWUWS_WCDGSDE	SD2_CLK	SD2_CLK	O/T	SPI1_CLK	I/O/T	SE_GPIO11	I/O/T	GPIO116	I/O/T
AY6	PDWUWS_WCDGSDE	SD2_CMD	SD2_CMD	I/O/T	SPI1_DI	I/O/T	SE_GPIO12	I/O/T	GPIO117	I/O/T

Ball No	Cell Name	Ball name	Function0	Type	Function1	Type	Function2	Type	Function3	Type
AW5	PDWUWS WCDGSDE	SD2_D0	SD2_D0	I/O/T	SPI1_D O	I/O/T	SE_GPI O13	I/O/T	GPIO1 18	I/O/T
AW7	PDWUWS WCDGSDE	SD2_D1	SD2_D1	I/O/T	SE_GPI O7	I/O/T			GPIO1 19	I/O/T
AR9	PDWUWS WCDGSDE	SD2_D2	SD2_D2	I/O/T	SE_GPI O8	I/O/T	EXTINT 8(G1)	I/O/T	GPIO1 20	I/O/T
AT8	PDWUWS WCDGSDE	SD2_D3	SD2_D3	I/O/T	SPI1_CS N	I/O/T	SE_GPI O9	I/O/T	GPIO1 21	I/O/T
F20	SPSCBC2_ 8X	SCL2	SCL2	I/O/T					GPIO1 22	I/O/T
G21	SPSCBC2_ 8X	SDA2	SDA2	I/O/T					GPIO1 23	I/O/T
B30	SPSCBC2_ 8X_W	KEYOU T1	KEYOU T1	I/O/T	EXTINT 12	I/O/T	CM4_G PIO6	I/O/T	GPIO1 24	I/O/T
A31	SPSCBC2_ 8X_W	KEYOU T0	KEYOU T0	I/O/T	EXTINT 11	I/O/T	CM4_G PIO5	I/O/T	GPIO1 25	I/O/T
A29	SPSCBC2_ 8X_W	KEYOU T2	KEYOU T2	I/O/T	PWMB	I/O/T	CM4_G PIO7	I/O/T	GPIO1 26	I/O/T
A27	SPSCBC2_ 8X	EXTINT 9	EXTINT 9	I/O/T	CM4_G PIO0	I/O/T			GPIO1 27	I/O/T
B28	SPSCBC2_ 8X	EXTINT 10	EXTINT 10	I/O/T	CM4_G PIO1	I/O/T			GPIO1 28	I/O/T
W7	SPSCBC2_ 8X	SCL6	SCL6	I/O/T					GPIO1 29	I/O/T
V8	SPSCBC2_ 8X	SDA6	SDA6	I/O/T					GPIO1 30	I/O/T
V4	SPSCBC2_ 8X	VDSP_T DO	VDSP_T DO	O/T			ESE_GP IO0	I/O/T	GPIO1 31	I/O/T
V6	SPSCBC2_ 8X	VDSP_T DI	VDSP_T DI	I/O/T			ESE_GP IO1	I/O/T	GPIO1 32	I/O/T
Y6	SPSCBC2_ 8X	VDSP_T MS	VDSP_T MS	I/O/T			ESE_GP IO2	I/O/T	GPIO1 33	I/O/T
U3	SPSCBC2_ 8X	VDSP_T CK	VDSP_T CK	I			ESE_GP IO3	I/O/T	GPIO1 34	I/O/T
V2	SPSCBC2_ 8X	VDSP_ RTCK	VDSP_ RTCK	I/O/T			ESE_GP IO4	I/O/T	GPIO1 35	I/O/T
AY1 2	SPSCBC2_ 8X_W	SIM_DE T0	SIM_DE T0	I/O/T					GPIO1 36	I/O/T

Ball No	Cell Name	Ball name	Function0	Type	Function1	Type	Function2	Type	Function3	Type
AW13	SPSCBC2_8X_W	SIM_DET1	SIM_DET1	I/O/T					GPIO1_37	I/O/T
AU13	SPSCBC2_8X_W	TF_DET	TF_DET	I/O/T					GPIO1_38	I/O/T
AV14	SPSCBC2_8X	BAT_DET	BAT_DET	I/O/T					GPIO1_39	I/O/T
U1	SPSCBC2_8X	U2TXD	U2TXD	O			DBG_BUS26(G1)	I/O/T	GPIO1_40	I/O/T
T2	SPSCBC2_8X	U2RXD	U2RXD	I			DBG_BUS27(G1)	I/O/T	GPIO1_41	I/O/T
T4	SPSCBC2_8X	U1TXD	U1TXD	O			DBG_BUS24(G1)	I/O/T	GPIO1_42	I/O/T
R3	SPSCBC2_8X	U1RXD	U1RXD	I	WDRST	I/O/T	DBG_BUS25(G1)	I/O/T	GPIO1_43	I/O/T
M6	SPSCBC2_8X	MTCK_ARM	MTCK_ARM	I					GPIO1_44	I/O/T
L5	SPSCBC2_8X	MTMS_ARM	MTMS_ARM	I/O/T					GPIO1_45	I/O/T
P4	SPSCBC2_8X	AUD_DSP_TDO	AUD_DSP_TDO	I/O/T			C2K_DSP_TDO	I/O/T	GPIO1_46	I/O/T
P2	SPSCBC2_8X	AUD_DSP_RTC_K	AUD_DSP_RTC_K	O	SE_GPIO6	I/O/T	C2K_DSP_RTC_K	I/O/T	GPIO1_47	I/O/T
N1	SPSCBC2_8X	AUD_DSP_TDI	AUD_DSP_TDI	I/O/T			C2K_DSP_TDI	I/O/T	GPIO1_48	I/O/T
M2	SPSCBC2_8X	AUD_DSP_TCK	AUD_DSP_TCK	I	PWMD	I/O/T	C2K_DSP_TCK	I	GPIO1_49	I/O/T
R1	SPSCBC2_8X	AUD_DSP_TMS	AUD_DSP_TMS	I/O/T	SE_GPIO10	I/O/T	C2K_DSP_TMS	I/O/T	GPIO1_50	I/O/T
C27	SPSEBC2_24X	SD1_CMD	SD1_CMD	I/O/T					GPIO1_51	I/O/T
C29	SPSEBC2_24X	SD1_D0	SD1_D0	I/O/T	LCM1_RSTN	I/O/T			GPIO1_52	I/O/T
D28	SPSEBC2_24X	SD1_D1	SD1_D1	I/O/T	DSI1_TE	I/O/T			GPIO1_53	I/O/T

Ball No	Cell Name	Ball name	Function0	Type	Function1	Type	Function2	Type	Function3	Type
D26	SPSEBC2_24X	SD1_CLK	SD1_CLK	O/T	PCIE3_RST	I/O/T			GPIO1_54	I/O/T
C25	SPSEBC2_24X	SD1_D2	SD1_D2	I/O/T	PCIE3_CLKREQ	I/O/T			GPIO1_55	I/O/T
E27	SPSEBC2_24X	SD1_D3	SD1_D3	I/O/T	PCIE3_WAKE	I/O/T			GPIO1_56	I/O/T
B38	SPSCBC2_8X	CLK_AUX0	CLK_AUX0	O	PROBE_CLK	I/O/T			GPIO1_57	I/O/T
A37	SPSCBC2_8X	U0TXD	U0TXD	O					GPIO1_58	I/O/T
A35	SPSCBC2_8X	U0RXD	U0RXD	I					GPIO1_59	I/O/T
B36	SPSCBC2_8X	U0CTS	U0CTS	I					GPIO1_60	I/O/T
B34	SPSCBC2_8X	U0RTS	U0RTS	O					GPIO1_61	I/O/T
A41	SPSCBC2_8X	U4TXD	U4TXD	O					GPIO1_62	I/O/T
B42	SPSCBC2_8X	U4RXD	U4RXD	I					GPIO1_63	I/O/T
C41	SPSCBC2_8X	U4CTS	U4CTS	I					GPIO1_64	I/O/T
C39	SPSCBC2_8X	U4RTS	U4RTS	O					GPIO1_65	I/O/T
B32	SPSCBC2_8X	IIS0DI	IIS0DI	I					GPIO1_66	I/O/T
C31	SPSCBC2_8X	IIS0DO	IIS0DO	O/T					GPIO1_67	I/O/T
A33	SPSCBC2_8X	IIS0CLK	IIS0CLK	I/O/T					GPIO1_68	I/O/T
E31	SPSCBC2_8X	IIS0LRC_K	IIS0LRC_K	I/O/T					GPIO1_69	I/O/T
G27	SPSCBC2_8X	PWMC	PWMC	I/O/T	CM4_GPIO3(G1)	I/O/T			GPIO1_70	I/O/T
G29	SPSCBC2_8X	IIS4DO	IIS4DO	O/T					GPIO1_71	I/O/T

Ball No	Cell Name	Ball name	Function0	Type	Function1	Type	Function2	Type	Function3	Type
D30	SPSCBC2_8X	IIS4DI	IIS4DI	I					GPIO172	I/O/T
F28	SPSCBC2_8X	IIS4CLK	IIS4CLK	I/O/T					GPIO173	I/O/T
E29	SPSCBC2_8X	IIS4LRC	IIS4LRC	I/O/T	CM4_GPIO4(G1)	I/O/T			GPIO174	I/O/T
B40	SPSCBC2_8X	U5TXD	U5TXD	O	SCL5	I/O/T			GPIO175	I/O/T
A39	SPSCBC2_8X	U5RXD	U5RXD	I	SDA5	I/O/T			GPIO176	I/O/T
F24	SPSCBC2_8X	SPI2_CS_N	SPI2_CS_N	I/O/T	EXTINT5(G1)	I/O/T			GPIO177	I/O/T
F22	SPSCBC2_8X	SPI2_DO	SPI2_DO	I/O/T					GPIO178	I/O/T
E23	SPSCBC2_8X	SPI2_DI	SPI2_DI	I/O/T					GPIO179	I/O/T
G23	SPSCBC2_8X	SPI2_CLK	SPI2_CLK	I/O/T	EXTINT6(G1)	I/O/T			GPIO180	I/O/T
M4	SPSCBC2_8X	U7TXD	U7TXD	O			DBG_BUS30(G1)	I/O/T	GPIO181	I/O/T
L3	SPSCBC2_8X	U7RXD	U7RXD	I			DBG_BUS31(G1)	I/O/T	GPIO182	I/O/T
N3	SPSCBC2_8X	U6TXD	U6TXD	O	PLL_LOCK	I/O/T	DBG_BUS28(G1)	I/O/T	GPIO183	I/O/T
N5	SPSCBC2_8X	U6RXD	U6RXD	I	MGPLL_SWITCH	I/O/T	DBG_BUS29(G1)	I/O/T	GPIO184	I/O/T
R7	SPSCBC2_8X_W	DMIC_CLK0	DMIC_CLK0	O			IIS5DO	O/T	GPIO185	I/O/T
P6	SPSCBC2_8X_W	DMIC_DAT0	DMIC_DAT0	I/O/T			IIS5DI	I	GPIO186	I/O/T
T6	SPSCBC2_8X_W	DMIC_CLK1	DMIC_CLK1	O			IIS5CLK	I/O/T	GPIO187	I/O/T
U7	SPSCBC2_8X_W	DMIC_DAT1	DMIC_DAT1	I/O/T			IIS5LRC	I/O/T	GPIO188	I/O/T

Ball No	Cell Name	Ball name	Function0	Type	Function1	Type	Function2	Type	Function3	Type
U5	SPSCBC2_8X_W	DMIC_CLK2	DMIC_CLK2	O					GPIO189	I/O/T
R5	SPSCBC2_8X_W	DMIC_DAT2	DMIC_DAT2	I/O/T					GPIO190	I/O/T
AM6	SPSCBC2_8X	IIS3DI	IIS3DI	I	EXTINT14(G1)	I/O/T			GPIO191	I/O/T
AL7	SPSCBC2_8X	IIS3DO	IIS3DO	O/T	EXTINT15(G1)	I/O/T			GPIO192	I/O/T
AM8	SPSCBC2_8X	IIS3LRCK	IIS3LRCK	I/O/T	SDA7	I/O/T			GPIO193	I/O/T
AN7	SPSCBC2_8X	IIS3CLK	IIS3CLK	I/O/T	SCL7	I/O/T			GPIO194	I/O/T
BA3	SPSCBC2_8X	CLK_AUX2	CLK_AUX2	O					GPIO195	I/O/T
AU15	SPSCBC2_8X	CHG_TYPE	CHG_TYPE	I					GPIO196	I/O/T
AT18	SPSCBC2_8X	DNS_D0	DNS_D0	O	DNS_PGA_LVL_L	O			GPIO197	I/O/T
AV18	SPSCBC2_8X	DNS_D1	DNS_D1	O	DNS_PGA_LVL_R	O			GPIO198	I/O/T
BA21	SPSCBC2_8X	ADI_D	ADI_D	I/O						
AY22	SPSCBC2_8X	ADI_SCLK	ADI_SCLK	O						
AW17	SPSCBC2_8X	ANA_INT0	ANA_INT0	I						
AW19	SPSCBC2_8X	ANA_INT1	ANA_INT1	I						
AR21	SPSCBC2_8X	ANA_INT2	ANA_INT2	I						
AY16	SPSCBC2_8X	AUD_ADD0	AUD_ADD0	I	IIS2DI	I				
AU17	SPSCBC2_8X	AUD_ADD1	AUD_ADD1	I	AON_ADCSW	O				
AR17	SPSCBC2_8X	AUD_ADSYNC	AUD_ADSYNC	I						

Ball No	Cell Name	Ball name	Function0	Type	Function1	Type	Function2	Type	Function3	Type
AV16	SPSCBC2_8X	AUD_DAD0	AUD_DAD0	O	IIS2DO	O/T				
AY18	SPSCBC2_8X	AUD_DAD1	AUD_DAD1	O						
BA17	SPSCBC2_8X	AUD_DASYNC	AUD_DASYNC	O	IIS2LRC	I/O/T				
AT16	SPSCBC2_8X	AUD_SCLK	AUD_SCLK	O	IIS2CLK	O/T				
AT20	SPSCBC2_8X	CHIP_SLEEP	CHIP_SLEEP	O						
AY20	SPSCBC2_8X	CLK_32K	CLK_32K	I						
BA19	SPSCBC2_8X	DCDCCPU0_EN	DCDCCPU0_EN	O						
AU19	SPSCBC2_8X	DCDCCPU1_EN	DCDCCPU1_EN	O						
AP16	SPSCBC2_8X	DCDCCPU2_EN	DCDCCPU2_EN	O						
AP20	SPSCBC2_8X	EXT_RST_B	EXT_RST_B	I						
BA15	SPSCBC2_8X	XTL_EN0	XTL_EN0	O						
AW15	SPSCBC2_8X	XTL_EN1	XTL_EN1	O						
AR19	SPSCBC2_8X	XTL_EN2	XTL_EN2	O						
AT22	SPSCBC2_8X_W	SSUSB_SW	SSUSB_SW	I						
D38	SPSCBC2_8X	ESE_GPIO	ESE_GPIO	I/O/T						

3.6 Pin Status and GPIOs

Ball No	Ball Name	GPIO	At Reset		After Reset		Power Domain
			H/L/Hiz	Pin State	H/L/Hiz/X	Pin State	
BA13	SIMCLK0	GPIO1	L	INPUT(WPD)	L	INPUT(WPD)	VSIM0
BA11	SIMDAT0	GPIO2	L	INPUT(WPD)	L	INPUT(WPD)	VSIM0
AY10	SIMRST0	GPIO3	L	INPUT(WPD)	L	INPUT(WPD)	VSIM0
AP10	SIMCLK1	GPIO4	L	INPUT(WPD)	L	INPUT(WPD)	VSIM1
AT10	SIMDAT1	GPIO5	L	INPUT(WPD)	L	INPUT(WPD)	VSIM1
AR11	SIMRST1	GPIO6	L	INPUT(WPD)	L	INPUT(WPD)	VSIM1
E25	KEYIN2	GPIO9	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AP22	PWMA	GPIO10	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AV22	LCM0_RSTN	GPIO11	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AW21	DSI0_TE	GPIO12	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AV20	EXTINT0	GPIO13	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AU21	EXTINT1	GPIO14	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AR23	SDA3	GPIO15	H	INPUT(WPU)	H	INPUT(WPU)	VIO1V8
AP24	SCL3	GPIO16	H	INPUT(WPU)	H	INPUT(WPU)	VIO1V8
AV8	SD0_D3	GPIO17	L	INPUT(WPD)	X	INPUT(WPD)	VSD0
AU9	SD0_D2	GPIO18	L	INPUT(WPD)	X	INPUT(WPD)	VSD0
AW9	SD0_CMD	GPIO19	L	INPUT(WPD)	X	INPUT(WPD)	VSD0
BA9	SD0_D0	GPIO20	L	INPUT(WPD)	X	INPUT(WPD)	VSD0
BA7	SD0_D1	GPIO21	L	INPUT(WPD)	X	INPUT(WPD)	VSD0
AY8	SD0_CLK	GPIO22	L	INPUT(WPD)	X	INPUT(WPD)	VSD0
AT40	EMMC_D0	GPIO23	H	INPUT(WPU)	H	INPUT(WPU)	VEMMC IO
AR41	EMMC_CMD	GPIO24	H	INPUT(WPU)	H	INPUT(WPU)	VEMMC IO
AY42	EMMC_D6	GPIO25	H	INPUT(WPU)	H	INPUT(WPU)	VEMMC IO
AW41	EMMC_D7	GPIO26	H	INPUT(WPU)	H	INPUT(WPU)	VEMMC IO
AU43	EMMC_CLK	GPIO27	L	OUTPUT	L	OUTPUT	VEMMC IO

Ball No	Ball Name	GPIO	At Reset		After Reset		Power Domain
			H/L/Hiz	Pin State	H/L/Hiz/X	Pin State	
AR39	EMMC_D5	GPIO28	H	INPUT(WPU)	H	INPUT(WPU)	VEMMC IO
AV42	EMMC_D4	GPIO29	H	INPUT(WPU)	H	INPUT(WPU)	VEMMC IO
AT42	EMMC_DS	GPIO30	L	INPUT(WPD)	L	INPUT(WPD)	VEMMC IO
AP38	EMMC_D3	GPIO31	H	INPUT(WPU)	H	INPUT(WPU)	VEMMC IO
AR43	EMMC_RST	GPIO32	H	OUTPUT	H	OUTPUT	VEMMC IO
AW43	EMMC_D1	GPIO33	H	INPUT(WPU)	H	INPUT(WPU)	VEMMC IO
AU41	EMMC_D2	GPIO34	H	INPUT(WPU)	H	INPUT(WPU)	VEMMC IO
A17	CMMCLK0	GPIO35	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
C17	CMMCLK1	GPIO36	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
B18	CMMCLK2	GPIO37	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
E17	CMMCLK3	GPIO38	L	INPUT(WPD)	X	INPUT(WPD)	VIO1V8
J3	CMRST0	GPIO39	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
K6	CMRST1	GPIO40	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
J7	CMRST2	GPIO41	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
L1	CMRST3	GPIO42	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
J5	CMRST4	GPIO43	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
K2	CMRST5	GPIO44	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
G17	CMPD0	GPIO45	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
K4	CMPD1	GPIO46	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
D18	CMPD2	GPIO47	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
H16	CMPD3	GPIO48	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
F18	CMPD4	GPIO49	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
H18	CMPD5	GPIO50	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
A19	SCL0	GPIO51	H	INPUT(WPU)	H	INPUT(WPU)	VIO1V8
B20	SDA0	GPIO52	H	INPUT(WPU)	H	INPUT(WPU)	VIO1V8

Ball No	Ball Name	GPIO	At Reset		After Reset		Power Domain
			H/L/Hiz	Pin State	H/L/Hiz/X	Pin State	
C19	SCL1	GPIO53	H	INPUT(WPU)	H	INPUT(WPU)	VIO1V8
E19	SDA1	GPIO54	H	INPUT(WPU)	H	INPUT(WPU)	VIO1V8
C21	SCL8	GPIO55	H	INPUT(WPU)	H	INPUT(WPU)	VIO1V8
D20	SDA8	GPIO56	H	INPUT(WPU)	H	INPUT(WPU)	VIO1V8
H20	SCL9	GPIO57	H	INPUT(WPU)	H	INPUT(WPU)	VIO1V8
G19	SDA9	GPIO58	H	INPUT(WPU)	H	INPUT(WPU)	VIO1V8
A21	SPI3_CSN	GPIO59	H	INPUT(WPU)	H	INPUT(WPU)	VIO1V8
B22	SPI3_CLK	GPIO60	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
E21	SPI3_DI	GPIO61	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
D22	SPI3_DO	GPIO62	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AJ5	RFSPi_SDA0	GPIO63	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AH4	RFSPi_SDA1	GPIO64	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AJ3	RFSPi_SDA2	GPIO65	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AK2	RFSPi_SDA3	GPIO66	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AJ1	RFSPi_SCK	GPIO67	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AH2	RFSPi_SEN	GPIO68	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AP6	RFCTL0(G0)	GPIO69	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AP8	RFCTL1(G0)	GPIO70	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AR7	RFCTL2(G0)	GPIO71	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AU7	RFCTL3(G0)	GPIO72	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AR5	RFCTL4(G0)	GPIO73	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AT6	RFCTL5(G0)	GPIO74	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AU5	RFCTL6(G0)	GPIO75	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AT4	RFCTL7(G0)	GPIO76	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AU3	RFCTL8(G0)	GPIO77	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AV4	RFCTL9(G0)	GPIO78	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AY4	RFCTL10(G0)	GPIO79	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AY2	RFCTL11(G0)	GPIO80	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AW3	RFCTL12(G0)	GPIO81	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AW1	RFCTL13(G0)	GPIO82	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8

Ball No	Ball Name	GPIO	At Reset		After Reset		Power Domain
			H/L/Hiz	Pin State	H/L/Hiz/X	Pin State	
AV2	RFCTL14(G0)	GPIO83	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AU1	RFCTL15(G0)	GPIO84	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AL5	RFFE0_SCK	GPIO85	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AM4	RFFE0_SDA	GPIO86	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AK6	RFFE1_SCK	GPIO87	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AK8	RFFE1_SDA	GPIO88	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AK4	RFFE2_SCK	GPIO89	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AL3	RFFE2_SDA	GPIO90	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AN5	RFFE3_SCK	GPIO91	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AN3	RFFE3_SDA	GPIO92	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AP4	RFFE4_SCK	GPIO93	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AR3	RFFE4_SDA	GPIO94	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AT2	RFFE5_SCK	GPIO95	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AR1	RFFE5_SDA	GPIO96	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AN1	RFFE6_SCK	GPIO97	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AP2	RFFE6_SDA	GPIO98	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AM2	RFFE7_SCK	GPIO99	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AL1	RFFE7_SDA	GPIO100	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AH6	LVDSRF_AD DAC0_ON	GPIO101	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AJ7	LVDSRF_AD DAC1_ON	GPIO102	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AH8	LVDSRF_AD DAC2_ON	GPIO103	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AG7	LVDSRF_AD DAC3_ON	GPIO104	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
A23	SPI0_CSN	GPIO105	H	INPUT(WPU)	H	INPUT(WPU)	VIO1V8
B24	SPI0_DO	GPIO106	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
C23	SPI0_DI	GPIO107	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
A25	SPI0_CLK	GPIO108	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AV12	IIS1DI	GPIO109	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8

Ball No	Ball Name	GPIO	At Reset		After Reset		Power Domain
			H/L/Hiz	Pin State	H/L/Hiz/X	Pin State	
AW11	IIS1DO	GPIO110	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AV10	IIS1CLK	GPIO111	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AU11	IIS1LRCK	GPIO112	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AT12	SCL4	GPIO113	H	INPUT(WPU)	H	INPUT(WPU)	VIO1V8
AP12	SDA4	GPIO114	H	INPUT(WPU)	H	INPUT(WPU)	VIO1V8
AY14	CLK_AUX1	GPIO115	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
BA5	SD2_CLK	GPIO116	L	INPUT(WPD)	L	INPUT(WPD)	VSD2
AY6	SD2_CMD	GPIO117	L	INPUT(WPD)	L	INPUT(WPD)	VSD2
AW5	SD2_D0	GPIO118	L	INPUT(WPD)	L	INPUT(WPD)	VSD2
AW7	SD2_D1	GPIO119	L	INPUT(WPD)	L	INPUT(WPD)	VSD2
AR9	SD2_D2	GPIO120	L	INPUT(WPD)	L	INPUT(WPD)	VSD2
AT8	SD2_D3	GPIO121	L	INPUT(WPD)	L	INPUT(WPD)	VSD2
F20	SCL2	GPIO122	H	INPUT(WPU)	H	INPUT(WPU)	VIO1V8
G21	SDA2	GPIO123	H	INPUT(WPU)	H	INPUT(WPU)	VIO1V8
B30	KEYOUT1	GPIO124	L	WPD	L	WPD	VIO1V8
A31	KEYOUT0	GPIO125	L	OUTPUT	L	OUTPUT	VIO1V8
A29	KEYOUT2	GPIO126	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
A27	EXTINT9	GPIO127	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
B28	EXTINT10	GPIO128	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
W7	SCL6	GPIO129	H	INPUT(WPU)	H	INPUT(WPU)	VIO1V8
V8	SDA6	GPIO130	H	INPUT(WPU)	H	INPUT(WPU)	VIO1V8
V4	VDSP_TDO	GPIO131	Hiz	Hiz	Hiz	Hiz	VIO1V8
V6	VDSP_TDI	GPIO132	H	INPUT(WPU)	H	INPUT(WPU)	VIO1V8
Y6	VDSP_TMS	GPIO133	H	INPUT(WPU)	H	INPUT(WPU)	VIO1V8
U3	VDSP_TCK	GPIO134	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
V2	VDSP_RTCK	GPIO135	L	OUTPUT	L	OUTPUT	VIO1V8
AY12	SIM_DET0	GPIO136	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AW13	SIM_DET1	GPIO137	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AU13	TF_DET	GPIO138	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AV14	BAT_DET	GPIO139	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8

Ball No	Ball Name	GPIO	At Reset		After Reset		Power Domain
			H/L/Hiz	Pin State	H/L/Hiz/X	Pin State	
U1	U2TXD	GPIO140	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
T2	U2RXD	GPIO141	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
T4	U1TXD	GPIO142	H	INPUT(WPU)	X	OUTPUT	VIO1V8
R3	U1RXD	GPIO143	H	INPUT(WPU)	H	INPUT(WPU)	VIO1V8
M6	MTCK_ARM	GPIO144	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
L5	MTMS_ARM	GPIO145	H	INPUT(WPU)	H	INPUT(WPU)	VIO1V8
P4	AUD_DSP_TD O	GPIO146	Hiz	Hiz	Hiz	Hiz	VIO1V8
P2	AUD_DSP_RT CK	GPIO147	L	OUTPUT	L	OUTPUT	VIO1V8
N1	AUD_DSP_TD I	GPIO148	H	INPUT(WPU)	H	INPUT(WPU)	VIO1V8
M2	AUD_DSP_TC K	GPIO149	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
R1	AUD_DSP_T MS	GPIO150	H	INPUT(WPU)	H	INPUT(WPU)	VIO1V8
C27	SD1_CMD	GPIO151	H	INPUT(WPU)	H	INPUT(WPU)	VIO1V8
C29	SD1_D0	GPIO152	H	INPUT(WPU)	H	INPUT(WPU)	VIO1V8
D28	SD1_D1	GPIO153	H	INPUT(WPU)	H	INPUT(WPU)	VIO1V8
D26	SD1_CLK	GPIO154	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
C25	SD1_D2	GPIO155	H	INPUT(WPU)	H	INPUT(WPU)	VIO1V8
E27	SD1_D3	GPIO156	H	INPUT(WPU)	H	INPUT(WPU)	VIO1V8
B38	CLK_AUX0	GPIO157	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
A37	U0TXD	GPIO158	H	OUTPUT	X	OUTPUT	VIO1V8
A35	U0RXD	GPIO159	H	INPUT(WPU)	H	INPUT(WPU)	VIO1V8
B36	U0CTS	GPIO160	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
B34	U0RTS	GPIO161	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
A41	U4TXD	GPIO162	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
B42	U4RXD	GPIO163	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
C41	U4CTS	GPIO164	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
C39	U4RTS	GPIO165	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
B32	IIS0DI	GPIO166	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8

Ball No	Ball Name	GPIO	At Reset		After Reset		Power Domain
			H/L/Hiz	Pin State	H/L/Hiz/X	Pin State	
C31	IIS0DO	GPIO167	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
A33	IIS0CLK	GPIO168	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
E31	IIS0LRCK	GPIO169	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
G27	PWMC	GPIO170	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
G29	IIS4DO	GPIO171	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
D30	IIS4DI	GPIO172	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
F28	IIS4CLK	GPIO173	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
E29	IIS4LRCK	GPIO174	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
B40	U5TXD	GPIO175	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
A39	U5RXD	GPIO176	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
F24	SPI2_CSN	GPIO177	H	INPUT(WPU)	H	INPUT(WPU)	VIO1V8
F22	SPI2_DO	GPIO178	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
E23	SPI2_DI	GPIO179	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
G23	SPI2_CLK	GPIO180	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
M4	U7TXD	GPIO181	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
L3	U7RXD	GPIO182	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
N3	U6TXD	GPIO183	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
N5	U6RXD	GPIO184	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
R7	DMIC_CLK0	GPIO185	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
P6	DMIC_DAT0	GPIO186	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
T6	DMIC_CLK1	GPIO187	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
U7	DMIC_DAT1	GPIO188	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
U5	DMIC_CLK2	GPIO189	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
R5	DMIC_DAT2	GPIO190	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AM6	IIS3DI	GPIO191	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AL7	IIS3DO	GPIO192	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AM8	IIS3LRCK	GPIO193	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AN7	IIS3CLK	GPIO194	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
BA3	CLK_AUX2	GPIO195	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AU15	CHG_TYPE	GPIO196	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8

Ball No	Ball Name	GPIO	At Reset		After Reset		Power Domain
			H/L/Hiz	Pin State	H/L/Hiz/X	Pin State	
AT18	DNS_D0	GPIO197	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8
AV18	DNS_D1	GPIO198	L	INPUT(WPD)	L	INPUT(WPD)	VIO1V8

NOTE

“X” means that the GPIO logic could be changed in the “After Reset”.

3.7 Pin Register Descriptions

See *UIS7885 Pin List and Pin Register Instructions*.

3.8 Strapping Pins

Ball name	Signal name in boot code	Value	Descriptions	Default value
U1TXD	USB_DLOAD_EN	0	USB download(no time out)	1(ie=1,wpu)
		1	UART download	
ADI_SCLK	ARM_BOOT_MD1	00	Reserved	11(ie=1,wpu) (MSB=MD1, LSB=MD0)
		01	UFS boot	
ADI_D	ARM_BOOT_MD0	10	SD boot(no time out)	
		11	eMMC 6.5MHz	
AUD_DAD0	ARM_BOOT_MD3	0	NON-KEY Download(With timeout) mode	1(ie=1,wpu)
		1	Normal mode	

Press key	Descriptions
Only Keyin0	USB download with timeout
Keyin0 + Keyin1	SD boot with timeout

3.9 Pad Information

The digital IO have 6 types:

- SPSCBC2_8X_VL/HL
- SPSCBC2_8X_W_VL/HL
- SPSCBC2_8X_SR_VL/HL
- SPSEBC2_24X_HL/VL
- PDWUWSWCDGSDE_V/H
- SPSEBC2_SWP_HL/VL

3.9.1 SPSCBC2_8X_VL/HL

SPSCBC2_8X_VL/HL is the normal pad used usually.

Truth Table of SPSCBC2_8X_VL/HL Pad is shown in [Table 3-4](#). Input function C = PAD&IE.

Table 3-4 Truth Table of SPSCBC2_8X_VL/HL Pad

Function	A	PAD	C	IE	OE	WPU	WPUS	WPDO	WPDI
Input	X	0	0	1	0	0	X	0	0
	X	1	1	1	0	0	X	0	0
Output	0	0	0	0	1	0	X	0	0
	1	1	0	0	1	0	X	0	0
Inout	0	0	0	1	1	0	X	0	0
	1	1	1	1	1	0	X	0	0
Tri State	X	HZ	0	0	0	0	X	0	0
	X	HZ	0	0	0	0	X	0	0
Pull up	X	1	0	0	0	WPU/WPUS: [00, 01, 10, 11]=[0, 4.7K, 20K, 1.8K]		0	0
Pull Down	X	0	0	0	0	0	X	1	0
IO Power-down	X	X	1	X	X	X	X	X	1

Driver Strength Select Function of SPSCBC2_8X_VL/HL Pad is shown in [Table 3-5](#).

VDDIO = 1.8V, All Corner (SS) @0.2 × VDDIO; 0.8 × VDDIO.

Table 3-5 Driver Strength Select Function of SPSCBC2_8X_VL/HL Pad

DRV[1]	DRV[0]	Driver Current
0	0	2mA
0	1	4mA
1	0	6mA
1	1	8mA

DC Parameters of SPSCBC2_8X_VL/HL Pad are shown in [Table 3-6](#). VDDIO = 1.8V.

Table 3-6 DC Parameter of SPSCBC2_8X_VL/HL Pad

Parameter	Description	SS	TT	FF
VT+	Schmitt trig. Low to High threshold point(1.8V)	X	>1.0V	X
VT-	Schmitt trig. High to low threshold point(1.8V)	X	>0.8V	X
RPU	Pull-up resistor(1.8V)	20k/4.7k/1.8k ±50%	20k/4.7k/1.8k ±50%	20k/4.7k/1.8k ±50%
RPD	Pull-down resistor(1.8V)	50k ± 70%	50k ± 70%	50k ± 70%
Ist	Standby Current	<2μA	<2μA	<2μA
IoZ	Tri-state output leakage current @Vo=3V or 0V	<2μA	<2μA	<2μA

AC Parameters of SPSCBC2_8X_VL/HL Pad are shown in [Table 3-7](#).

The data simulation is SS corner, Cloding = 0.3pF, Cpad = 30pF/15pF, VDDIO = 1.8V, Temp = 125°C.

Table 3-7 AC Parameter of SPSCBC2_8X_VL/HL Pad

Delay Path		Rise Time (ns)	Fall Time (ns)	Rise Delay (ns)	Fall Delay (ns)
PAD->C		0.193	0.198	0.437	0.327
A-PAD	2mA	4.911	4.766	3.379	3.54
	4mA	2.569	2.491	2.452	2.51
	6mA	1.829	1.778	2.168	2.193
	8mA	2.609	2.548	2.491	2.521

3.9.2 SPSCBC2_8X_SR_VL/HL

Base on the SPSCBC2_8X_VL/HL, improved the slew rate performance.

Truth Table of SPSCBC2_8X_SR_VL/HL Pad is shown in [Table 3-8](#). Input function C=PAD&IE.

Table 3-8 Truth Table of SPSCBC2_8X_SR_VL/HL Pad

Function	A	PAD	C	IE	OE	WPU	WPUS	WPDO	WPDI
Input	X	0	0	1	0	0	X	0	0
	X	1	1	1	0	0	X	0	0
Output	0	0	0	0	1	0	X	0	0

Function	A	PAD	C	IE	OE	WPU	WPUS	WPDO	WPDI
	1	1	0	0	1	0	X	0	0
Inout	0	0	0	1	1	0	X	0	0
	1	1	1	1	1	0	X	0	0
Tri State	X	HZ	0	0	0	0	X	0	0
	X	HZ	0	0	0	0	X	0	0
Pull up	X	1	0	0	0	WPU/WPUS: [00, 01, 10, 11]=[0, 4.7K, 20K, 1.8K]		0	0
Pull Down	X	0	0	0	0	0	X	1	0
IO Power-down	X	X	1	X	X	X	X	X	1

Driver Strength Select Function of SPSCBC2_8X_SR_VL/HL Pad is shown in [Table 3-9](#).

VDDIO = 1.8V, All Corner (SS) @0.2 × VDDIO; 0.8 × VDDIO.

Table 3-9 Driver Strength Select Function of SPSCBC2_8X_SR_VL/HL Pad

DRV[1]	DRV[0]	Driver Current
0	0	2mA
0	1	4mA
1	0	6mA
1	1	8mA

Table 3-10 DC Parameter of SPSCBC2_8X_SR_VL/HL Pad

Parameter	Description	SS	TT	FF
VT+	Schmitt trig. Low to High threshold point(1.8V)	X	1.20V	X
VT-	Schmitt trig. High to low threshold point(1.8V)	X	0.64V	X
RPU	Pull-up resistor(1.8V)		WPU/WPUS: [00, 01, 10, 11]=[0, 4.7K, 20K, 1.8K]	
RPD	Pull-down resistor(1.8V)	33K	50K	66K
Iavg	Active Current@100MHz,FF Corner,30pF loading	X	X	4mA
Ist	Standby Current	X	X	<0.1μA

Parameter	Description	SS	TT	FF
IoZ	Tri-state output leakage current @Vo=3V or 0V	X	X	<0.1μA

AC Parameters of SPSCBC2_8X_SR_VL/HL Pad are shown in [Table 3-11](#).

The data simulation is SS corner, Cloding = 0.3pF, Cpad = 30pF/15pF, VDDIO = 1.8V, Temp = 125°C.

Table 3-11 AC Parameter of SPSCBC2_8X_SR_VL/HL Pad

Delay Path		Rise Time(ns)	Fall Time(ns)	Rise Delay(ns)	Fall Delay(ns)
PAD->C		0.193	0.198	0.437	0.327
A-PAD	2mA	4.911	4.766	3.379	3.54
	4mA	2.569	2.491	2.452	2.51
	6mA	1.829	1.778	2.168	2.193
	8mA	2.609	2.548	2.491	2.521

3.9.3 SPSCBC2_8X_W_VL/HL

Base on the SPSCBC2_8X_VL/HL, enlarge the ESD area for Keypad and etc 8KV ESD performance.

Truth Table of SPSCBC2_8X_W_VL/HL Pad is shown in [Table 3-12](#). Input function C = PAD&IE

Table 3-12 Truth Table of SPSCBC2_8X_W_VL/HL Pad

Function	A	PAD	C	IE	OE	WPU	WPUS	WPDO	WPDI
Input	X	0	0	1	0	0	X	0	0
	X	1	1	1	0	0	X	0	0
Output	0	0	0	0	1	0	X	0	0
	1	1	0	0	1	0	X	0	0
Inout	0	0	0	1	1	0	X	0	0
	1	1	1	1	1	0	X	0	0
Tri State	X	HZ	0	0	0	0	X	0	0
	X	HZ	0	0	0	0	X	0	0
Pull up	X	1	0	0	0	Soft Pull-Up select WPU/WPUS: [00, 01, 10, 11]=[0, 4.7K, 20K, 1.8K]		0	0
Pull Down	X	0	0	0	0	0	X	1	0
IO Power-down	X	X	1	X	X	X	X	X	1

Driver Strength Select Function of SPSCBC2_8X_W_VL/HL Pad is shown in [Table 3-13](#).

VDDIO = 1.8V, All Corner (SS) @0.2 × VDDIO; 0.8 × VDDIO.

Table 3-13 Driver Strength Select Function of SPSCBC2_8X_W_VL/HL Pad

DRV[1]	DRV[0]	Driver Current
0	0	2mA
0	1	4mA
1	0	6mA
1	1	8mA

Table 3-14 DC Parameter of SPSCBC2_8X_W_VL/HL Pad

Parameter	Description	SS	TT	FF
V _{T+}	Schmitt trig. Low to High threshold point(1.2V)	X	0.80V	X
V _{T-}	Schmitt trig. High to low threshold point(1.2V)	X	0.39V	X
V _{T+}	Schmitt trig. Low to High threshold point(1.8V)	X	1.20V	X
V _{T-}	Schmitt trig. High to low threshold point(1.8V)	X	0.64V	X
RPU	Pull-up resistor(1.8V)		WPU/WPUS: [00, 01, 10, 11]=[0, 4.7K, 20K, 1.8K]	
RPD	Pull-down resistor(1.8V)	33K	50K	66K
Iavg	Active Current@100MHz,FF Corner,30pF loading	X	X	4mA
Ist	Standby Current	X	X	<0.1μA
IoZ	Tri-state output leakage current @Vo=3V or 0V	X	X	<0.1μA

AC Parameters of SPSCBC2_8X_W_VL/HL Pad are shown in [Table 3-15](#).

The data simulation is SS corner, Cloding = 0.3pF, Cpad = 15pF, VDDIO = 1.8V, Temp = 125°C.

Table 3-15 AC Parameter of SPSCBC2_8X_W_VL/HL Pad

Delay Path		Rise Time(ns)	Fall Time(ns)	Rise Delay(ns)	Fall Delay(ns)
PAD->C		0.002	0.0014	0.118	0.084
A-PAD	100Ω/9mA	1.45	1.54	0.21	0.144
	66Ω/13mA	1.01	1.24	0.2	0.134
	50Ω/18mA	0.74	0.89	0.187	0.127
	40Ω/23mA	0.658	0.72	0.17	0.125

Delay Path		Rise Time(ns)	Fall Time(ns)	Rise Delay(ns)	Fall Delay(ns)
	33Ω/27mA	0.57	0.63	0.164	0.141
	20Ω/45mA	0.344	0.356	0.193	0.155

3.9.4 SPSEBC2_24X_VL/HL

SPSEBC2_24X_VL/HL IO PAD is design for eMMC. The function is a managed memory capable of storing code and data. It is specifically designed for mobile devices. The eMMC is intended to offer the performance and features required by mobile devices while maintaining low power consumption. The eMMC device contains features that support high throughput for large data transfers and performance for small random data more commonly found in code usage. It also contains many security features.

Truth Table of SPSEBC2_24X_VL/HL Pad is shown in [Table 3-16](#). Input function C = PAD&IE.

Table 3-16 Truth Table of SPSEBC2_24X_VL/HL Pad

Function	A	PAD	C	IE	OE	WPU	WPUS	WPDO	WPDI
Input	X	0	0	1	0	0	X	0	0
	X	1	1	1	0	0	X	0	0
Output	0	0	0	0	1	0	X	0	0
	1	1	0	0	1	0	X	0	0
Inout	0	0	0	1	1	0	X	0	0
	1	1	1	1	1	0	X	0	0
Tri State	X	HZ	0	0	0	0	X	0	0
	X	HZ	0	0	0	0	X	0	0
Pull up	X	1	0	0	0	Soft Pull-Up select WPU/WPUS: [00, 01, 10, 11]=[0, 4.7K, 20K, 1.8K]		0	0
Pull Down	X	0	0	0	0	0	X	1	0
IO Power-down	X	X	1	X	X	X	X	X	1

Table 3-17 Driver Strength Select Function of SPSEBC2_24X_VL/HL Pad

DRV[2]	DRV[1]	DRV[0]	Driver Strength I
0	0	0	5mA
0	0	1	10mA
0	1	0	15mA
0	1	1	20mA

DRV[2]	DRV[1]	DRV[0]	Driver Strength I
1	0	0	25mA
1	0	1	30mA
1	1	0	35mA
1	1	1	40mA

Table 3-18 DC Parameter of SPSEBC2_24X_VL/HL Pad

Parameter	Description	SS	TT	FF
VT+	Schmitt trig. Low to High threshold point(1.8V)	X	>1.0V	X
VT-	Schmitt trig. High to low threshold point(1.8V)	X	>0.8V	X
RPU	Pull-up resistor(1.8V)	20K/4.7K/1.5K $\pm$ 50%	20K/4.7K/1.5K $\pm$ 50%	20K/4.7K/1.5K $\pm$ 50%
RPD	Pull-down resistor(1.8V)	50k $\pm$ 70%	50k $\pm$ 70%	50k $\pm$ 70%
Ist	Standby Current	<2 μ A	<2 μ A	<2 μ A
IoZ	Tri-state output leakage current @Vo=1.8V or 0V	<2 μ A	<2 μ A	<2 μ A

AC Parameters of SPSEBC2_24X_VL/HL Pad are shown in [Table 3-19](#).

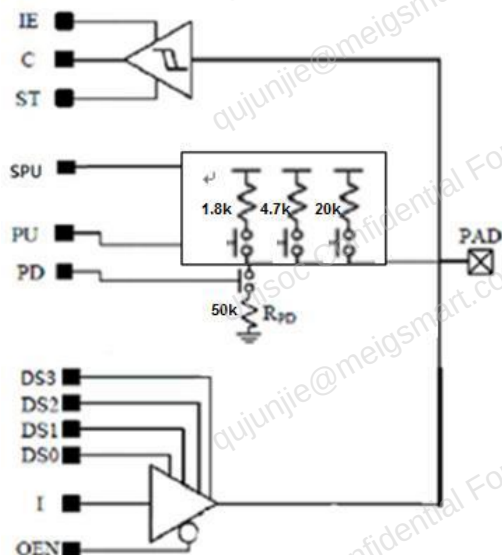
The data simulation is SS corner, clk = 208MHz, Cloding = 0.2pF, VDDIO = 1.8V, Temp = 125°C.

Table 3-19 AC Parameter of SPSEBC2_24X_VL/HL Pad

Delay Path (mA)		Raising Time (ns)	Fall Time (ns)	Raising Delay (ns)	Fall Delay (ns)
PAD -> C		0.51	0.736	0.162	0.153
A-> PAD	5	5.594	14.547	9.386	25.06
	10	3.031	6.634	3.774	10.637
	15	2.365	4.457	2.315	6.246
	20	1.976	3.443	1.666	4.31
	25	2.029	2.956	1.349	3.275
	30	1.83	2.57	1.131	2.631
	35	1.728	2.311	0.969	2.198
	40	1.615	2.111	0.858	1.886

3.9.5 PDWUWSWCDGSDE_V/H

Figure 3-2 Diagram of PDWUWSWCDGSDE_V/H Pad



Truth Table of PDWUWSWCDGSDE_V/H Pad is shown in [Table 3-20](#).

Input function C = PAD&IE, Output function: PAD = A&OE.

Table 3-20 Truth Table of PDWUWSWCDGSDE_V/H Pad

INPUT													OUTPUT		Remark
MS	OEN	I	DS0	DS1	DS2	DS3	PU	SPU	PD	ST	IE	PAD	PAD	C	
1	0	0	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	-	0	0	1.8V UD operation
1	0	1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0	-	1	0	
1	0	1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	1	-	1	1	
0	0	0	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	-	0	0	3.0V OD operation
0	0	1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0	-	1	0	
0	0	1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	1	-	1	1	
0/1	1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0	-	0	Input mode
0/1	1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0	1	-	0	
0/1	1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	1	1	-	1	
0/1	1	0/1	0/1	0/1	0/1	0/1	0	0	0	0/1	0	Z	-	0	
0/1	1	0/1	0/1	0/1	0/1	0/1	0	0	0	0/1	1	Z	-	X	
0/1	1	0/1	0/1	0/1	0/1	0/1	1	0	1	0/1	0/1	N.A.	-	N.A.	Forbidden
0/1	1	0/1	0/1	0/1	0/1	0/1	0	1	1	0/1	0/1		-		
0/1	1	0/1	0/1	0/1	0/1	0/1	1	0	0	0/1	0	Z	H	0	20k weak pull
0/1	1	0/1	0/1	0/1	0/1	0/1	1	0	0	0/1	1	Z	H	1	
0/1	1	0/1	0/1	0/1	0/1	0/1	0	1	0	0/1	0	Z	H	0	4.7k strong pull
0/1	1	0/1	0/1	0/1	0/1	0/1	0	1	0	0/1	1	Z	H	1	
0/1	1	0/1	0/1	0/1	0/1	0/1	1	1	0	0/1	0	Z	H	0	1.8k strong pull
0/1	1	0/1	0/1	0/1	0/1	0/1	1	1	0	0/1	1	Z	H	1	
0/1	1	0/1	0/1	0/1	0/1	0/1	0	0	1	0/1	0/1	Z	L	0	Pull down

Driver Strength Select Function of PDWUWSWCDGSDE_V/H Pad is shown in [Table 3-21](#).

1.8V mode: All Corner (SS) @ $0.5 \times VDDIO$.

Table 3-21 Driver Strength Select Function of PDWUWSWCDGSDE_V/H Pad

DS3	DS2	DS1	DS0	Driver Current
0	0	0	0	2.4mA
0	0	0	1	5.5mA
0	0	1	0	7.5mA
0	0	1	1	10.4mA
0	1	0	0	12.4mA
0	1	0	1	15mA
0	1	1	0	17.4mA
0	1	1	1	20mA
1	0	0	0	24.6mA
1	0	0	1	27.8mA
1	0	1	0	29.6mA
1	0	1	1	32mA
1	1	0	0	34.2mA
1	1	0	1	37mA
1	1	1	0	39mA
1	1	1	1	42mA

3.9.6 SPSEBC2_SWP_HL/VL

SPSEBC2_SWP_VL/HL IO PAD is design for eSE SWP pin. The IO character is the same with SPSEBC2_4X_VL/HL IO, the only difference is the PD resistor, SWP IO is 2.25K ($\pm 20\%$) for 800 μ A current detect and 24X IO is 50K ($\pm 70\%$).

- 1.62V ~ 1.98V VDDIO supply operation
- 0.72V ~ 0.88V VDD supply operation
- Operating junction temperature range: -40°C ~ 125°C
- Single wire protocol Bi-direction Pad support Voltage input or Voltage output or Current output.
- Minimum Signal bit duration: 1 μ s.
- Total area(width $\times$ height): 60 μ m $\times$ 335 μ m

Pin List

Table 3-22 SWPIO pin list1

Pin Name	Voltage Level	Description
VDD	0.75V	Core Power Supply.
VSS	0V	Core Ground Supply.
VDDIO	1.875V	IO post driver Power Supply.
VSSIO	0V	IO post driver Ground Supply.

Table 3-23 SWPIO pin list2

Pin Name	Default	Description
WPDO		SWP S2 current signal output control: 1'b0 current output=low(0μA) 1'b1 current output=high(800μA)
WPDI	1'b0	<i>If it is not used, the pin is floating.</i>
WPU	1'b0	Soft Pull-Up Enable, Active=1
WPUS	1'b0	Soft Pull-Up select signal.0 for big res, and 1 for small res.
A		Input pin from digital to SWPIO PAD
PAD		Bonding PAD pin
PADI		PAD signal after the second protection resistor. Can be used for analog signal.
C		Output pin to digital from SWPIO PAD
DRV[2]	1'b0	Driver Strength Select pin, bit 2
DRV[1]	1'b0	Driver Strength Select pin, bit 1
DRV[0]	1'b0	Driver Strength Select pin, bit 0
OE	1'b0	Output Enable, Active=1. Single wire protocol do not need voltage output function, it can be used in debug mode.
IE	1'b1	Input Enable, Active=1
SE	1'b1	Schmitt trigger Input Enable, Active=1
POC	1'b0	SWPIO output isolation when VDD is powered off, Active=1 tie SWPIO output to high/low; POC=0 SWPIO work in normal mode.(POC come from SPVDD2POC_VL_base)

Pin Name	Default	Description
SWP_BG_ITRIM[3:0]	4'b0111	trim SWP output current: 4'b0000 => -46.6% 4'b0001 => -40% ... 4'b0110 => -6.7% 4'b0111(default) => + 0% 4'b1000 => + 6.7% ... 4'b1110 => + 46.6% 4'b1111 => + 53.4%
SWP_BG_RBIAS_MODE	1'b0	SWP BG rbias mode: 1'b0 turn off rbias mode(default); 1'b1 turn on rbias mode;
SWP_BG_TEST_EN	1'b0	SWP BG output reference test path: 1'b0 turn off output reference test path(default); 1'b1 turn on output reference test path;
SWP_PG_PD	1'b0	SWP BG power up/down: 1'b0 power up swp bg(default); 1'b1 power down swp bg;

Function Description

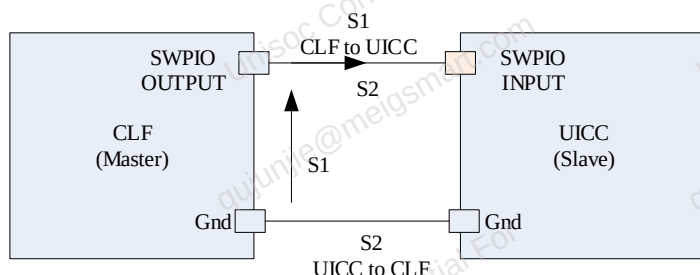
Table 3-24 SWPIO cell list

Name	Area	Description
SPSCBC2_SWP_VL/HL	60μm × 335μm	Bi-direction Pad with Enable Controlled Pull-Up Resistor. Programmable Driver Strength 2mA/4mA/6mA/8mA. No tolerance function. The Schmitt trigger input function can be controlled by the SE signal. SE=1, Schmitt input enable. SE=0 Schmitt input disable. Support 800μA current output controlled by WPDO.

Principle of the Single Wire Protocol

The SWP interface is a bit oriented, point to point communication protocol between a UICC and a contactless front-end(CLF) as shown in figure below.

Figure 3-3 SWP data transmission



The principle of the Single Wire Protocol is based on the transmission of digital information in full duplex mode:

1. The signal S1 is transmitted by a digital modulation (L or H) in the voltage domain.
2. The signal S2 is transmitted by a digital modulation (L or H) in the current domain.

Electrical Specifications

Table 3-25 SWPIO electrical Specifications

Item	Parameter	Specification			Unit
		Min	Typical	Max	
VDDIO	SWPIO supply (Class C)	1.62		1.98	V
VDD	Digital supply	0.72		0.88	V
Icc	Average current over 1ms			5	mA
Temperature	Junction temperature	-40	30	125	
Area	IP in die 60μm × 335μm		0.02		mm ²
V _{OH}	Master SWP S1 (PAD) output high voltage when $I_{Lmin} \leq S2 \leq I_{Hmax}$ (Class C)	1.377		1.98	V
V _{OL}	Master SWP S1 (PAD) output Low voltage when $-20\mu A \leq S2 \leq 0\mu A$ (Class C)	0		0.297	V
V _{IH}	Slave SWP S1(PAD) input high voltage (Class C)	1.134		2.28	V
V _{IL}	Slave SWP S1(PAD) input low voltage (Class C)	-0.3		0.495	V
I _H	Slave SWP S2(PAD) output high current when $V_{IHmin} \leq S1 \leq I_{Hmax}$	600		1000	μA
I _L	Slave SWP S2(PAD) output low current when $V_{IHmin} \leq S1 \leq I_{Hmax}$	0		20	μA
tf	Fall time of S1 when $C_{load} \leq 10pF$, $T < 5\mu s$	5ns		0.05×T	
	Fall time of S1 when $C_{load} \leq 10pF$, $T > 5\mu s$	5ns		250ns	
tr	Rise time of S1 when $C_{load} \leq 10pF$, $T < 5\mu s$	5ns		0.05×T	

Item	Parameter	Specification			Unit
		Min	Typical	Max	
	Rise time of S1 when $C_{load} \leq 10pF$, $T > 5\mu s$	5ns		250ns	
T_{H1}	Duration of the state H for coding a logical 1 of S1	$0.7 \times T$	$0.75 \times T$	$0.8 \times T$	
T_{H1}	Duration of the state H for coding a logical 0 of S1	$0.2 \times T$	$0.25 \times T$	$0.3 \times T$	
T	Default bit duration	1		5	μs
	Extended bit duration	0.59		10	μs
I _q	S1 mode at $T=1\mu s$				
	S2 mode at $T=1\mu s$				

Figure 3-4 SWPIO Voltage definition for the signal S1

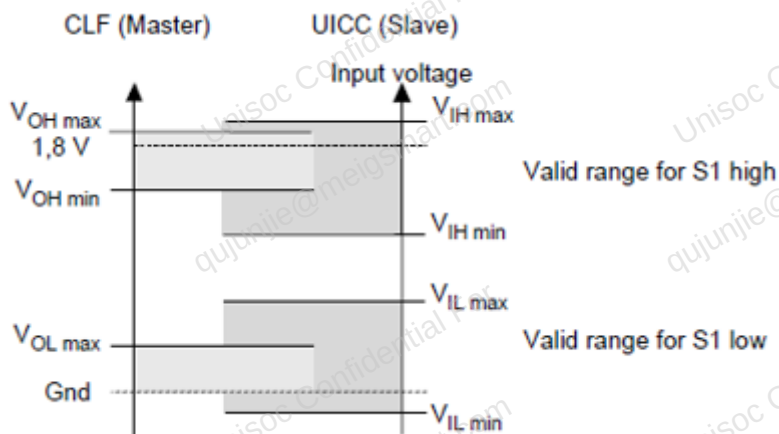
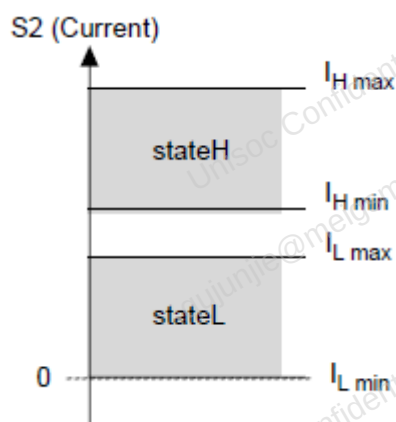


Figure 3-5 SWPIO Current definition for the signal S2



The delay path inside SWPIO: (SS corner, Cloding = 0.3pF, Cpad = 30pF/15pF, VDDIO = 1.875V, Temp = 125°C).

Table 3-26 SWPIO delay path

Delay Path		Rise Time(ns)	Fall Time(ns)	Rise Delay(ns)	Fall Delay(ns)
PAD->C		0.193	0.198	0.437	0.327
A->PAD	2mA	4.911	4.766	3.379	3.54
	4mA	2.569	2.491	2.452	2.51
	6mA	1.829	1.778	2.168	2.193
	8mA	2.609	2.548	2.491	2.521
WPDO->PAD		2.51	4.138	3.09	3.97

SPSCBC2_SWP_VL/HL setting

Bi-direction Pad with Enable Controlled Pull-Up Resistor. Programmable Driver Strength. No tolerance function. The Schmitt trigger input function can be control by the SE signal. SE = 1, Schmitt input enable. SE = 0 Schmitt input disable. Support 800μA current output controlled by WPDO.

Figure 3-6 SWPIO diagram

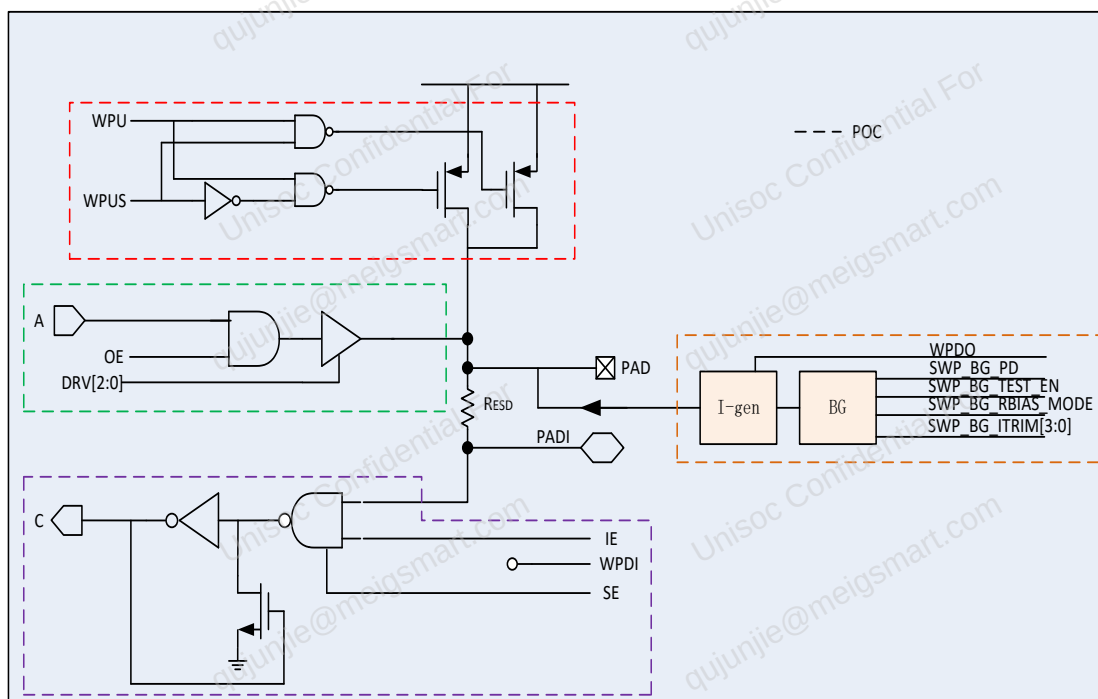
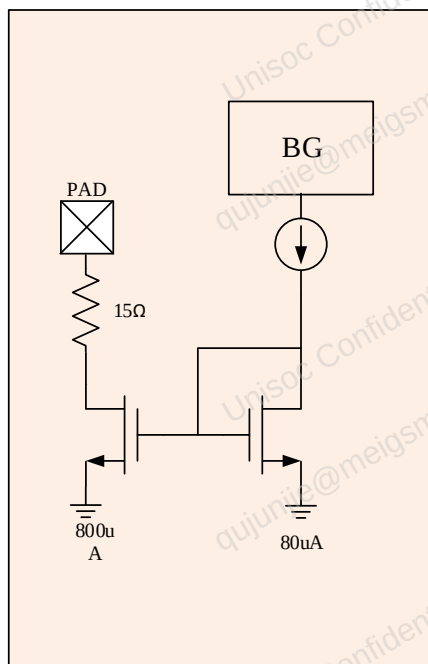


Figure 3-7 SWPIO current generation



Truth Table: Input function C = PAD&IE.

- Output function: PAD = A&OE.
- Output current function: PAD = WPDO&(~SWP_BG_PD).

Table 3-27 SWPIO application setting

Function	A	PAD	C	IE	OE	WPU	WPUS	WPDO	WPDI	SWP_PG_PD
Input voltage	X	0	0	1	0	0	X	0	0	1
	X	1	1	1	0	0	X	0	0	1
Output Voltage	0	0	0	0	1	0	X	0	0	1
	1	1	0	0	1	0	X	0	0	1
Output current	X	1(I _L)	0	0	0	0	X	0	0	0
	X	1(I _H)	0	0	0	0	X	0	1	0
Tri State	X	HZ	0	0	0	0	X	0	0	1
	X	HZ	0	0	0	0	X	0	0	1
Pull up	X	1	0	0	0	1	0: 20K(1.8V) 1: 4.7K(1.8V)	0	0	1

Table 3-28 Driver Strength Select Function of SPSEBC2_SWP_HL/VL Pad

DRV[2]	DRV[1]	DRV[0]	Driver Current
0	0	0	2mA
0	0	1	4mA
0	1	0	6mA
0	1	1	8mA

4 Electrical Specification

4.1 DC Specification

4.1.1 Absolute Maximum Ratings

The functionality is subject to the absolute maximum/minimum values listed in [Table 4-1](#). Do not exceed these parameters or the part may be damaged permanently. Operation at absolute maximum ratings is not guaranteed.

Table 4-1 Absolute maximum ratings

Symbol	Parameter	Min	Max	Unit
V _{DI_18V}	Input voltage on 1.8V domain digital input(IO)	-0.3	1.98	V
V _{DI_3V}	Input voltage on 3V domain digital input(IO)	-0.3	3.3	V
V _{AI_18V}	Input voltage on 1.8V domain analog input	-0.3	1.98	V
V _{AI_12V}	Input voltage on 1.2V domain analog input	-0.3	1.3	V
V _{max, ESD}	HBM (Human Body Model) (IO)		2000	V
I _{max, DC}	Maximum DC Input current for any non-supply pin(IO)		5	mA
T _{storage}	Storage temperature	-40	+125	°C
T _c	Case temperature(SoC)	-20	+85	°C
T _{efuse}	Fuse programming temperature(SoC)	-30	+85	°C

4.1.2 Recommended Operating Conditions

The recommended operating conditions are list in [Table 4-2](#).

(Recommended voltages limits refers to die-level internal voltages)

Table 4-2 Recommended operating conditions

Symbol	Parameter	Min	Typical	Max	Unit
VDDCORE	Power supply for Core domain	0.585 0.675	0.65(deep sleep) 0.75	0.715 0.825	V
VDDCPU_LIT	Ananke supply voltage	0.585		1.045	V
VDDCPU_MID	Prometheus supply voltage	0.585		1.045	V
VDDCPU_BIG	Prometheus supply voltage	0.675		1.1	V

Symbol	Parameter	Min	Typical	Max	Unit
VDDAI	AI supply voltage	0.585		0.825	V
VDDMM	MM supply voltage	0.585		0.825	V
VDDGPU	GPU supply voltage	0.585		0.825	V
VDDMODEM	Modem supply voltage	0.585		0.825	V
VDDSRAM	SRAM supply voltage	0.675	0.75	0.825	V
VDDSRAM_CPU_LIT	Power supply for dual rail SRAM of Little core of CPU	0.675		1.045	V
VDDSRAM_CPU_MID	Power supply for dual rail SRAM of Mid core of CPU	0.675		1.045	V
AVDD1V8_BB	Baseband supply voltage	1.62	1.8	1.98	V
VDDPUB	Power supply for LPDDR4X controller	0.585		0.825	V
VDDMEM	Power supply for LPDDR4X core power	1.06	1.1	1.17	V
VDDMEMQ	Power supply for LPDDR4X IO	0.57	0.6	0.65	V
AVDD1V8_EFU	EFUSE 1.8V analog power supply	1.62	1.8	1.98	V
AVDD1V8_MPLLBIG	MPLLBIG 1.8V analog power supply	1.62	1.8	1.98	V
AVDD1V8_MPLLLIT	MPLLLIT 1.8V analog power supply	1.62	1.8	1.98	V
AVDD1V8_MPLLMID	MPLLMID 1.8V analog power supply	1.62	1.8	1.98	V
AVDD1V8_DPLL	DDR PLL supply voltage	1.62	1.8	1.98	V
AVDD1V8_PCIEPLL	PCIEPLL 1.8V analog power supply	1.62	1.8	1.98	V
AVDD1V8_USBPLL	USB3.1 PLL 1.8V analog power supply	1.62	1.8	1.98	V
AVDD1V8_DSI	MIPI DSI supply voltage	1.62	1.8	1.98	V
AVDD1V2_DSI	MIPI DSI supply voltage	1.08	1.2	1.32	V
AVDD1V8_CSI	MIPI CSI supply voltage	1.62	1.8	1.98	V
AVDD1V2_CSI	MIPI CSI supply voltage	1.08	1.2	1.32	V
AVDD1V8_LVDSRF	LVDSRF supply voltage	1.62	1.8	1.98	V
AVDD1V2_LVDSRF	LVDSRF supply voltage	1.08	1.2	1.32	V
AVDD1V2_USB	USB3.1 1.2V analog power supply	1.08	1.2	1.32	V
AVDD1V8_USB	USB PHY supply voltage	1.62	1.8	1.98	V
AVDD3V3_USB	USB PHY supply voltage	2.7	3.3	3.63	V
VDD0V75_PCIE	PCIe 0.75V digital power supply	0.675	0.75	0.825	V

Symbol	Parameter	Min	Typical	Max	Unit
VDD0V75_UFS	UFS 0.75V digital power supply	0.675	0.75	0.825	V
VDD0V75_USB	USB3.1 0.75V digital power supply	0.675	0.75	0.825	V
AVDD1V2_PCIE	PCIe 1.2V analog power supply	1.08	1.2	1.32	V
AVDD1V2_UFS	UFS 1.2V analog power supply	1.08	1.2	1.32	V
VDD1V8_ISE	ISE 1.8V digital power supply	1.62	1.8	1.98	V

4.1.3 ESD Characteristics

The ESD characteristics are shown in [Table 4-3](#).

Table 4-3 ESD characteristics

Symbol	Parameter	Condition	Value	Unit
HBM	Human body model	ANSI/ESDA/JEDEC JS-001	±2000	V
CDM	Charged-device model	ANSI/ESDA/JEDEC JS-002	±500	V
LU	Latch UP current	JESD78E	±100	mA

4.1.4 DC Characteristics

The typical core voltage (VDDCORE) is 0.75V and the digital I/O supply (VIO1V8) is typically at 1.875V. The power pins should be connected with a decoupling capacitor to ground (VSS).

For the following table, $T_{amb} = -30^{\circ}\text{C}$ to $+60^{\circ}\text{C}$, $VSS = 0\text{V}$ (ground), and all voltages are measured with respect to VSS, unless otherwise specified.

Table 4-4 DC Characteristics

Symbol	Parameter	Conditions	Min	Typical	Max	Unit
Digital supply voltage: pins VIO1V8						
VIO1V8	Digital supply voltage		1.62		1.98	V
Digital supply voltage: pins VSIM0						
VSIM0	Digital supply voltage		1.62		3.3	V
Digital supply voltage: pins VSIM1						
VSIM1	Digital supply voltage		1.62		3.3	V
Digital supply voltage: pins VSD0						
VSD0	Digital supply voltage		1.62		3.3	V
Digital supply voltage: pins VSD2						

Symbol	Parameter	Conditions	Min	Typical	Max	Unit
VSD2	Digital supply voltage		1.62		3.3	V
Digital supply voltage: pins VDDMEM/BDDMEMQ(DDR)						
VDDMEM	Digital supply voltage		1.06	1.1	1.17	V
VDDMEMQ	IO supply voltage		0.57	0.6	0.65	V
Digital supply voltage: pins VEMMCIO						
VEMMCIO	Digital supply voltage		1.62		1.98	V
Digital supply voltage: internal VDDCORE(SoC)						
VDDCORE	Digital core supply voltage		0.675	0.75	0.825	V
	Sleep mode digital core supply voltage		0.585	0.65	0.715	V
Digital input(IO)						
V _{IL}	Input voltage LOW-level		0		0.3 VDDIO	V
V _{IH}	Input voltage HIGH-level		0.7 VDDIO		VDDIO	V
I _{LI}	Input leakage current			±2		μA
Digital output(IO)						
V _{OL}	Output voltage LOW-level	At I _{sink} = 2,4,6,8 mA (programmable)	0		0.1 VDDIO	V
V _{OH}	Output voltage HIGH-level	At I _{source} = 2,4,6,8 mA (programmable)	0.9 VDDIO		VDDIO	V
Analog supply voltage						
AVDD1V8_BB	Baseband supply voltage		1.62	1.8	1.98	V
VDDPUB	Power supply for LPDDR4X controller		0.585		0.825	V
VDDMEM	Power supply for LPDDR4X core power		0.99	1.1	1.21	V
VDDMEMQ	Power supply for LPDDR4X IO		0.57	0.6	0.65	V
AVDD1V8_EFU	EFUSE 1.8V analog power supply		1.62	1.8	1.98	V
AVDD1V8_MPLLBIG	MPLLBIG 1.8V analog power supply		1.62	1.8	1.98	V

Symbol	Parameter	Conditions	Min	Typical	Max	Unit
AVDD1V8_MPLLLIT	MPLLLIT 1.8V analog power supply		1.62	1.8	1.98	V
AVDD1V8_MPLLMID	MPLLMID 1.8V analog power supply		1.62	1.8	1.98	V
AVDD1V8_DPLL	DDR pll supply voltage		1.62	1.8	1.98	V
AVDD1V8_PCIEPLL	PCIEPLL 1.8V analog power supply		1.62	1.8	1.98	V
AVDD1V8_USBPLL	USB3.1 PLL 1.8V analog power supply		1.62	1.8	1.98	V
AVDD1V8_DSI	MIPI DSI supply voltage		1.62	1.8	1.98	V
AVDD1V2_DSI	MIPI DSI supply voltage		1.08	1.2	1.32	V
AVDD1V8_CSI	MIPI CSI supply voltage		1.62	1.8	1.98	V
AVDD1V2_CSI	MIPI CSI supply voltage		1.08	1.2	1.32	V
AVDD1V8_LVDSRF	LVDSRF supply voltage		1.62	1.8	1.98	V
AVDD1V2_LVDSRF	LVDSRF supply voltage		1.08	1.2	1.32	V
AVDD1V8_USB	USB PHY supply voltage		1.7	1.8	1.98	V
AVDD3V3_USB	USB PHY supply voltage		3.0	3.3	3.6	V
VDD0V75_PCIE	PCIe 0.75V digital power supply		0.675	0.75	0.825	V
VDD0V75_UFS	UFS 0.75V digital power supply		0.675	0.75	0.825	V
VDD0V75_USB	USB3.1 0.75V digital power supply		0.675	0.75	0.825	V
AVDD1V2_PCIE	PCIe 1.2V analog power supply		1.08	1.2	1.32	V
AVDD1V2_UFS	UFS 1.2V analog power supply		1.08	1.2	1.32	V
AVDD1V2_USB	USB3.1 1.2V analog power supply		1.08	1.2	1.32	V

4.2 AC Characteristics

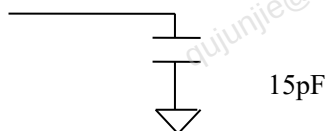
A pin's AC characteristics include input and output capacitance, which determine loading for external drivers or other load analysis. The AC characteristics also include a de-rating factor, which indicates how much faster or slower the AC timings get with different loads.

Table 4-5 Standard input, output and I/O pin AC characteristics

Symbol	Parameters	Min	Typical	Max	Unit
C_{in}	Input capacitance, all standard input and IO pins			3.5	pF
C_{load}	Output capacitance, all standard output and IO pins			30	pF
T_{dr}	Output de-rating falling edge on all standard output and I/O pins, from 15pF load		0.075		ns/pF

NOTE

The AC specifications are tested with a 15pF load as indicated

Figure 4-1 Test circuit of an I/O pin


The output capacitance and de-rating falling edge are measured under the condition of maximum driving strength: 24 mA @ 1.8 V.

For the following tables, $T_c = -20^{\circ}\text{C}$ to $+85^{\circ}\text{C}$, $V_{SS} = 0\text{ V}$ (ground), and all voltages are measured with respect to V_{SS} , unless otherwise specified.

Table 4-6 System Clock AC characteristics

Symbol	Parameter	Conditions	Min	Typical	Max	Unit
f_{mclk}	Master clock frequency			52		MHz
C_I	Input capacitance			30		pF
Master clock input						
V_{pp}	Sine wave peak-to-peak swing	Loading = 30pF	0.7		1.8	V
Real-time clock input						
f_{rtclk}	Real time clock frequency			32		kHz
$\Delta f/f_{rtclk}$	Frequency tolerance	Standard deviation	-80		80	ppm
δ_{rtclk}	Clock duty cycle		35		65	%

4.3 Performance Specifications

4.3.1 LVDSRF

The LVDS interface is designed for transmitting and receiving data between RF chip and BB chip. The recommended line rate is 1.04Gbps.

4.3.1.1 DC Specifications

Table 4-7 LVDS TX DC specifications

Parameter	Conditions/Comments	Min	Typical	Max	Unit
Vod	Transmit Differential output voltage magnitude	190	200	210	mV
$\Delta Vod $	Change in Differential output voltage magnitude between logic states			25	mV
VCMTX	Steady-state common-mode output voltage	180	200	220	mV
$\Delta VCMTX$	Changes in steady-state common-mode output voltage between logic states			25	mV
Zos	Single-ended output impedance		50		Ω

Table 4-8 LVDS RX DC specifications

Parameter	Conditions/Comments	Min	Typical	Max	Unit
VIDTH	Differential input high voltage threshold		50		mV
VIDTL	Differential input low voltage threshold	-50			mV
VCMRX_DC	Input common mode voltage		200		mV
ZID	Differential input impedance (before calibration)	80	100	125	Ω

4.3.1.2 AC Specifications

Table 4-9 LVDS TX AC specifications

Parameter	Conditions/Comments	Min	Typical	Max	Unit
Cnear	Near-end capacitor to ground		2	3	pF
Cfar	Near-end capacitor to ground		2	3	pF
tr	Differential output signal rise time (20%-80%)			0.1	UI
tf	Differential output signal fall time (80%-20%)			0.1	UI
$\Delta VCMTX$	Changes in common-mode output voltage when random bit stream in		50	100	mVpp

Table 4-10 LVDS RX AC specifications

Parameter	Conditions/Comments	Min	Typical	Max	Unit
$\Delta V_{CMRX}(HF)$	Common mode interference beyond 450MHz			200	mVpp
$\Delta V_{CMRX}(LF)$	Common mode interference between 50MHz and 450MHz	-50		50	mVpp
CCM	Common mode termination		20		pF

4.3.2 SOC Phase-Locked- Loop (PLL)

Table 4-11 Performance of the phase-locked loop

Parameter	Conditions/Comments	Min	Typical	Max	Unit
Input reference	All PLLs except DeskewPLL		26		MHz
Input reference	DeskewPLL	100	500	800/1066.5	MHz
Frequency Range	PIXELPLL	1600	2673	3200	MHz
	GPLL	800	858	1800	MHz
	AUDPLL	800	1228.88	1600	MHz
	VDSPPLL	800	1014	1600	MHz
	AIPLL	800	988	1600	MHz
	PHYCPR8PLL	800	1222	1222	MHz
	CPLL	1022.36	1022.36	1022.36	MHz
	MPLL_BIG	1200	2600	3200	MHz
	MPLL_MID	1200	2392	3200	MHz
	MPLL_LIT	1200	2002	2400	MHz
	MPLL_SCU	1000	1586	2000	MHz
	PSCPR8PLL	800	1222	1222	MHz
	TGPLL	1536	1536	1536	MHz
	V4NRPLL	2457.6	2457.6	2457.6	MHz
	LVDSPLL	1.456	2.912	4	GHz
Settling time	DPLL	100	1066	3200	MHz
	DeskewPLL	400	2000	4266	MHz
Sleep mode	Yes			200	μ s

4.3.3 Thermal Sensor

This is a temperature-to-digital converter using an on-chip bandgap temperature sensor. The overall accuracy is $\pm 3^{\circ}\text{C}$ (typical) from 0°C to 90°C .

This is suitable for multi-core smart phone applications requiring highly accurate temperature sensing to perform thermal control.

Table 4-12 Performance of the Thermal sensor

Parameter	Conditions/Comments	Min	Typical	Max	Unit
Power supply	Analog supply voltage	1.62	1.8	1.98	V
	Digital core supply voltage	0.675	0.75	0.825	V
Temperature resolution			0.26		$^{\circ}\text{C}$
Temperature accuracy	$0^{\circ}\text{C} \sim 90^{\circ}\text{C}$		± 3		$^{\circ}\text{C}$
	$-20^{\circ}\text{C} \sim 0^{\circ}\text{C}$, $90^{\circ}\text{C} \sim 125^{\circ}\text{C}$		± 5		$^{\circ}\text{C}$
Conversion time		24		32	32k Clock cycle

4.3.4 USB 2.0 PHY

The USB 2.0 PHY is fully compliant with USB 2.0 specifications. It's own PLL can generate 480MHz clock for its usage. Please refer to Universal Serial Bus Specification revision 2.0 at <http://www.usb.org/developers/docs>

Table 4-13 Main performance of the USB

Parameter	Conditions/Comments	Min	Typical	Max	Unit
Power supply	Analog 1.8V power supply	1.7	1.8	1.98	V
	Analog 3.3V power supply	3.0	3.3	3.6	V
	Digital core supply voltage	0.675	0.75	0.825	V
THSR	Rise time(10% ~ 90%)	500			ps
THSF	Fall time (90% ~ 10%)	500			ps
ZoutDRV(HS)	Driver output impedance	40.5	45	49.5	Ω
VCRS(FS)	Crossover voltage	1.3		2.0	V
TFRFM(FS)	Rise time and fall time matching	90	100	110	%
TFR(FS)	Rise time	4		20	ns
TFF(FS)	Fall time	4		20	ns
RPU1	Bus pull_up resistor1 on upstream port	900		1575	Ω
RPU2	Bus pull_up resistor2 on upstream port	525		1515	Ω
RPD	Bus pull_down resistor on downstream port	14.25k		24.8k	Ω
VDI	Differential input sensitivity	0.2			V

Parameter	Conditions/Comments	Min	Typical	Max	Unit
VCM	Differential input common mode range	0.8		2.5	V
TFST(FS)	Width of SE0 during differential transition			14	ns
Zout DRV(FS)	Driver output resistor	28		44	Ω
VCRS(LS)	Crossover voltage	1.3		2.0	V
TFRFM(LS)	Rise time and fall time matching	80	100	120	%
TFR(LS)	Rise time	75		200	ns
TFF(LS)	Fall time	75		200	ns
VHSSQ common range	Vhssq common input range	-50		500	mv
TFST(LS)	Width of SE0 during differential transition			210	ns

4.3.5 USB3.1/DP ComboPHY

The USB3.1/DP ComboPHY is fully compliant with USB3.1/DisplayPort1.4 specifications and PIPE version4.0 interface.

Table 4-14 Main performance of the USB3.1/DP

Parameter	Conditions/Comments	Min	Typical	Max	Unit
Power supply	Analog 1.8V power supply	1.7	1.8	1.98	V
	Analog low power supply	0.7	0.75	0.825	V
	Digital core supply voltage	0.7	0.75	0.825	V
Differential Insertion Loss	TX Differential Routes, at baudrate/2		-1.5		dB
Differential Return Loss	TX Differential Routes, at baudrate		-10		dB
Common Mode Return Loss	TX Differential Routes, at baudrate		-6		dB
Common Mode to differential loss	TX Differential Routes, at baudrate		-20		dB
Differential to common mode conversion	TX Differential Routes, at baudrate		-20		dB
Xtalk coupling(all attackers)	TX Differential Routes, at baudrate/2		-30		dB
Differential Impedance		90	100	110	Ω
Single Ended Impedance		45	50	55	Ω
Length variation (intra differential pair)	intra differential pair		± 0.250		mm
Length variation (between lanes)	between lanes		± 0.5		mm

Parameter	Conditions/Comments	Min	Typical	Max	Unit
Dj _{REFCLK} USB3.0	Reference clock deterministic jitter		12.0		ps
Dj _{REFCLK} USB3.1	Reference clock deterministic jitter		7.4		ps
Dj _{REFCLK} DP HBR2	Reference clock deterministic jitter		14.9		ps
Dj _{REFCLK} DP HBR3	Reference clock deterministic jitter		14.9		ps

4.3.6 MIPI CSI

MIPI CSI is fully compliant with MIPI CSI-2 V1.2 and D-PHY 1.2. Please refer specification at <http://www.mipi.org>

Table 4-15 Main performance of the MIPI CSI

Parameter	Conditions/Comments	Min	Typical	Max	Unit
Power supply	Analog 1.8V power supply Analog 1.2V power supply Digital core supply voltage	1.7 1.1 0.675	1.8 1.2 0.75	1.98 1.3 0.825	V V V
HS RX mode					
VCMRX	Common-mode voltage HS receiver mode	70		330	mV
Vidh	Differential input high threshold			70	mV
Vidtl	Differential input low threshold	-70			mV
Vihhs	Single-ended input high voltage			460	mV
Vilhs	Single-ended input low voltage	-40			mV
Zid	Differential input impedance	80		125	Ω
ΔVCMRX (HF)	Common-mode interference beyond 450MHz			100	mV
ΔVCMRX (LF)	Common-mode interference beyond 50~450MHz	-50		50	mV
LP RX mode					
VIH	Logic high	880			mV
VIL	Logic low			550	mV
VIL-ULPS	Logic low @ULP			300	mV
VHYST	Input hysteresis	25			mV
eSPIKE	Input pulse rejection			300	V•ps

Parameter	Conditions/Comments	Min	Typical	Max	Unit
TMIN-RX	Minimum pulse width response	50			ns
Data-Clock Timing Spec					
Tsetup(rx) @ 1.5Gbps		0.2			UI
Tsetup(rx) @ 1Gbps		0.15			UI
Thold(rx) @ 1.5Gbps		0.2			UI
Thold(rx) @ 1Gbps		0.15			UI

4.3.7 MIPI C-PHY CSI

MIPI C-PHY CSI is fully compliant with MIPI D-PHY V1.2 and C-PHY V1.2. Please refer specification at <http://www.mipi.org>

Parameter	Conditions/Comments	Min	Typical	Max	Unit
Power supply	Analog 1.8V power supply	1.7	1.8	1.98	V
	Analog 1.2V power supply	1.1	1.2	1.3	V
	Digital core supply voltage	0.675	0.75	0.825	V
HS RX mode					
VCMRX	Common-mode voltage HS receiver mode	95		390	mV
Width	Differential input high threshold			40	mV
Vidtl	Differential input low threshold	-40			mV
Vihhs	Single-ended input high voltage			535	mV
Vilhs	Single-ended input low voltage	-40			mV
Zid_ab Zid_bc Zid_ca	Differential input impedance	80	100	120	Ω
ΔZid	Differential input impedance mismatch			10	%
$\Delta VCMRX(HF)$	Common-mode interference beyond 450MHz			50	mV
$\Delta VCMRX(LF)$	Common-mode interference beyond 50~450MHz	-25		25	mV
LP RX mode					
VIH	Logic high	740			mV
VIL	Logic low			550	mV
VIL-ULPS	Logic low @ULP			300	mV
VHYST	Input hysteresis	25			mV
eSPIKE	Input pulse rejection			300	V•ps

Parameter	Conditions/Comments	Min	Typical	Max	Unit
TMIN-RX	Minimum pulse width response	20			ns
Timing Spec					
Receive timing budget		0.5			UI

4.3.8 MIPI DSI

MIPI DSI is fully compliant with MIPI DSI V1.2 and D-PHY 1.2. Please refer specification at <http://www.mipi.org>

Table 4-16 Main performance of the MIPI DSI

Parameter	Conditions/Comments	Min	Typical	Max	Unit
Power supply	Analog 1.8V power supply	1.7	1.8	1.98	V
	Analog 1.2V power supply	1.1	1.2	1.3	V
	Digital core supply voltage	0.675	0.75	0.825	V
HSTX mode					
Zos	Single ended output impedance	40	50	62.5	Ω
VCMTX	HS transmit static common-mode voltage	150	200	250	mV
$ \Delta VCMTX(1,0) $	VCMTX mismatch when output is Differential-1 or 0			5	mV
$ VOD $	HS transmit differential voltage	140	200	270	mV
$ \Delta VOD $	VOD mismatch when output is Differential-1 or 0			10	mV
VOHHS	HS output high voltage			360	mV
$\Delta VCMTX(HF)$	Common-level variations above 450MHz(RMS)			15	mV
$\Delta VCMTX(LF)$	Common-level variation between 50 ~450MHz(RMS)			25	mV
tr / tf	20% ~ 80% rise time and fall time(<1.5Gbps)	100p		0.35UI	ps
LPTX mode					
VOH	Thevenin output high level	1.1	1.2	1.3	V
VOL	Thevenin output low level	-50		50	mV
ZOLP	Output impedance of LP transmitter	110			Ω
$\Delta ZOLP(01,10)$	Single-ended output impedance mismatch driving opposite level			20	%
$\Delta ZOLP(00,11)$	Single-ended output impedance mismatch driving same level			5	%
Slew rate	Clload = 70pF	30		120	mV/ns
LPRX mode					

Parameter	Conditions/Comments	Min	Typical	Max	Unit
VIH	Logic high	740			mV
VIL	Logic low			550	mV
VHYST	Input hysteresis	25			mV
LPCD mode					
VIHCD	Logic high contention threshold	450			mV
VILCD	Logic low contention threshold			200	mV

4.3.9 MIPI MPHY4.1

MIPI MPHY is fully compliant with MIPI MPHY 4.1. Please refer specification at <http://www.mipi.org>

Table 4-17 Main performance of the MIPI MPHY

Parameter	Conditions/Comments	Min	Typical	Max	Unit
Power supply	Analog 1.8V power supply	1.7	1.8	1.98	V
	Analog 1.2V power supply	1.08	1.2	1.32	V
	Digital core supply voltage	0.72	0.8	0.88	V
VDIF_AC_LA_RT_TX	Large differential TX AC voltage amplitude in terminated case	140		250	mV
VDIF_DC_LA_NT_TX	Large differential TX DC voltage amplitude in unterminated case	320		480	mV
VDIF_AC_LA_NT_TX	Large differential TX AC voltage amplitude in unterminated case	280		500	mV
VDIF_DC_SA_RT_TX	Small differential TX DC voltage amplitude in terminated case	100		130	mV
VDIF_AC_SA_RT_TX	Small differential TX AC voltage amplitude in terminated case	80		140	mV
VDIF_DC_SA_NT_TX	Small differential TX DC voltage amplitude in unterminated case	200		260	mV
VDIF_AC_SA_NT_TX	Small differential TX AC voltage amplitude in unterminated case	160		280	mV
VCM_LA_TX	TX common mode voltage for large amplitude	160		260	mV
VCM_SA_TX	TX common mode voltage for small amplitude	80		190	mV
R_SE_TX	Single-ended transmitter output resistance, fixed calibration	40		60	Ω
HS TX mode					
TF_TX	Fall time	0.1			UI

Parameter	Conditions/Comments	Min	Typical	Max	Unit
TR_TX	Rise time	0.1			UI
TEYE_TX	Transmitter eye opening(HS-G1 and HS-G2)	0.2			UI
TEYE_HS_G3_TX	Transmitter eye opening(HS-G3)	0.55			UI
TSKEW_TX	Intra-differential skew	-0.06		0.06	UI
TPULSE_TX	Transmitter pulse width(HS-G1 for VDIF_DC_SA_RT_TX)	0.9			UI
ΔRSE_TX	Output resistance mismatch	-6		6	Ω
DJTX	Transmitter deterministic jitter			0.15	UI
TJTX	Transmitter total jitter			0.32	UI
STTJTX	Transmitter short term total jitter.			0.2	UI
fOFFSET_TX	Transmitter frequency offset	-2000		2000	ppm
LP TX mode					
TR_LS_TX	Rise time LS			0.2	UI_LS
TF_LS_TX	Fall time LS			0.2	UI_LS
Receiver Part					
Vdif_rt_rx	Differential RX voltage amplitude with termination	60		245	mV
Vdif_nt_rx	Differential RX voltage amplitude in untermination	120		490	mV
Vdif_acc_rx	Accumulated differential RX voltage amplitude for HS-GEAR1	40			mV
Vcm_rx	RX common-mode voltage	25		330	mV
R_Diff_RX	Differential input resistance	80		110	Ω
Teye_rx	High-speed Receiver eye opening	0.2			UI
Tpulse_rx	High-speed Receiver pulse width	0.8			UI
SJrx	High-speed Receiver sinusoidal jitter tolerance			0.35	UI
STSJrx	High-speed Receiver short term sinusoidal jitter tolerance			0.2	UI
TJrx	High-speed Receiver total jitter tolerance			0.52	UI
STTJrx	High-speed Receiver short term total jitter tolerance			0.3	UI
Vsq	Squelch exit voltage	50		140	mV

Parameter	Conditions/Comments	Min	Typical	Max	Unit
Vdif_sq	Squelch differential voltage			20	mV
Vcm_sq	Squelch common-mode voltage	0		330	mV
Tpulse_sq	Noise pulse width			20	ns
Tspace_sq	Noise pulse spacing	500			ns
Vint_sq	Peak interfere amplitude			200	mV
Fint_sq	Interfere frequency	500			MHz

4.3.10 PCIEGEN3

Table 4-18 Main performance of the PCIEGEN3 2.5GT/s

Parameter	Conditions/Comments	Min	Typical	Max	Unit
Power supply	Analog supply voltage	1.7	1.8	1.98	V
	Digital core supply voltage	0.72	0.8	0.88	V
UI	unit interval	400.002	400.004	400.005	ps
VTX-DIFF-PP	Differential p-p Tx voltage swing for full swing operation	810	868	897	mV
VTX-DIFF-PP-LOW	Differential p-p Tx voltage swing for low swing operation	417	449	467	mV
TTX-UTJ	Tx uncorrelated total jitter	8.74	10.3	14.7	ps
TTX-UDJDD	Tx uncorrelated deterministic jitter	1.67	2.56	6.36	ps
TTX-UTJ-SRIS	Tx Uncorrelated Total Jitter SRIS	16.6	20.1	21.7	ps
TX-DETECT-RX	TX detect RX functionality 1-Pass, 0-Fail	1.000	1.000	1.000	MHz
BWTX-PKG-PLL1	Tx PLL BW corresponding to PKG-Tx-PLL1	8.12	11.6	14.1	MHz
PKG-TX-PLL1	Tx PLL peaking corresponding to BWTX-PKG-PLL1	0.217	0.391	0.526	dB
VRX-IDLE-DET-DIFF-PP	Electrical idle detect threshold	105	118	138	mV

Table 4-19 Main performance of the PCIEGEN3 5GT/s

Parameter	Conditions/Comments	Min	Typical	Max	Unit
Power supply	Analog supply voltage	1.7	1.8	1.98	V
	Digital core supply voltage	0.72	0.8	0.88	V

Parameter	Conditions/Comments	Min	Typical	Max	Unit
UI	unit interval	200.001	200.002	200.003	ps
VTX-DIFF-PP	Differential p-p Tx voltage swing for full swing operation	813	863	895	mV
VTX-DIFF-PP-LOW	Differential p-p Tx voltage swing for low swing operation	420	450	466	mV
TTX-UTJ	Tx uncorrelated total jitter	7.41	8.86	14.1	ps
TTX-UDJDD	Tx uncorrelated deterministic jitter	0.939	1.91	3.21	ps
TTX-UPW-TJ	Total Uncorrelated Pulse Width Jitter (ps PP)	5.21	7.75	10.9	ps
TTX-UPW-DJDD	Deterministic DjDD Uncorrelated Pulse Width Jitter (ps PP)	1.20	2.80	5.16	ps
TTX-UTJ-SRIS	Tx Uncorrelated Total Jitter SRIS	7.275	8.316	10.873	ps
BWTX-PKG-PLL1	Tx PLL BW corresponding to PKG-Tx-PLL1	8.1	11.6	14.1	MHz
PKG-TX-PLL1	Tx PLL peaking corresponding to BWTX-PKG-PLL1	0.217	0.391	0.526	dB
VRX-IDLE-DET-DIFF-PP	Electrical idle detect threshold	93	108	126	mV

Table 4-20 Main performance of the PCIEGEN3 8GT/s

Parameter	Conditions/Comments	Min	Typical	Max	Unit
Power supply	Analog supply voltage	1.7	1.8	1.98	V
	Digital core supply voltage	0.72	0.8	0.88	V
UI	unit interval	125.001	125.001	125.002	ps
VTX-DIFF-PP	Differential p-p Tx voltage swing for full swing operation	811	861	890	mV
VTX-DIFF-PP-LOW	Differential p-p Tx voltage swing for low swing operation	413	446	464	mV
VTX-EIEOS-FS	Minimum voltage swing during EIEOS for full swing signaling	285	305	320	mV
VTX-EIEOS-RS	Minimum voltage swing during EIEOS for reduced swing signaling	276	296	309	mV
TTX-UTJ	Tx uncorrelated total jitter	5.86	6.79	8.67	ps
TTX-UDJDD	Tx uncorrelated deterministic jitter	0.309	0.780	1.31	ps
TTX-UPW-TJ	Total Uncorrelated Pulse Width Jitter (ps PP)	5.90	6.84	8.37	ps

Parameter	Conditions/Comments	Min	Typical	Max	Unit
TTX-UPW-DJDD	Deterministic DjDD Uncorrelated Pulse Width Jitter (ps PP)	0.843	1.46	2.11	ps
TTX-UTJ-SRIS	Tx Uncorrelated Total Jitter SRIS	9.894	11.685	13.496	ps
BWTX-PKG-PLL1	Tx PLL BW corresponding to PKG-Tx-PLL1	2.25	3.11	3.46	MHz
PKG-TX-PLL1	Tx PLL peaking corresponding to BWTX-PKG-PLL1	0.603	0.707	0.917	dB
VTX-BOOST-FS	maximum nominal Tx boost ratio for full swing	8.92	9.29	9.41	dB
VTX-BOOST-RS	maximum nominal Tx boost ratio for reduced swing	9.55	9.65	9.91	dB
VRX-IDLE-DET-DIFF-PP	Electrical idle detect threshold	93	108	125	mV

4.3.11 Automatic Power Control DAC

One general DAC is used to control power ramping and gain. It is a 14 bits D/A converters with a programmable-gain output driver. Special offset cancellation is applied in the DAC.

Table 4-21 Performance of the APC DAC

Parameter	Conditions/Comments	Min	Typical	Max	Unit
Power supply	Analog supply voltage(AVDD)	1.62	1.8	1.98	V
	Digital core supply voltage(DVDD)	0.675	0.75	0.825	V
Max Output range	PGA gain=1.3	0.03		1.3	V
	PGA gain=1.5	0.04		1.4	V
	PGA gain=2.0	0.05		1.5	V
N	Resolution of the DAC		14		Bit
Sample rate				1	MHz
PGA gain		1.3	1.5	2	
DNL	Differential nonlinearity		±1	±2	LSB
Setting time	4pF Capacitor load, full scale input, 1.5X PGA gain, bypass internal RC filter		1	2	μs
PSRR		40			dB

4.3.12 CLK26M Buffer

This CLK26M Buffer is used to generate 26MHz clock for PLL and other circuit blocks. It includes one 26M_buffer and one sine driver output of this CLK26M block can be set the sine wave.

Table 4-22 Performance of the 26MHz Buffer

Parameter	Conditions/Comments	Min	Typical	Max	Unit
Power supply	Analog supply voltage	1.62	1.8	1.98	V
	Digital core supply voltage	0.675	0.75	0.825	V
Vpp_in	Input sine wave peak to peak amplitude		325		mV
f_26	Output Clock Frequency		26		MHz
Dutycycle	Output Clock Dutycycle	45	50	55	%
f_sine	Output sine wave Frequency		26		MHz

4.3.13 RCO

Table 4-23 Performance of the RCO

Parameter	Conditions/Comments	Min	Typical	Max	Unit
Frequency range	RCO6M	5.94	6	6.06	MHz
	RCO60M	54	60	66	MHz
	RCO100M	90	100	110	MHz
	RCO150M	135	150	165	MHz

4.3.14 DDC

The DDC is used for reduction of on-die transient voltage droop with high-performance embedded circuit.

Table 4-24 Main performance of the DDC

Parameter	Conditions/Comments	Min	Typical	Max	Unit
AVDD18	Analog power supply(clean)	1.62	1.8	1.98	V
PVDD18	Line supply voltage input	1.62	1.8	1.98	V
DVDD	Digital power supply	0.675	0.75	0.825	V
Temp	Junction temperature range	-20	25	125	°C
IAVDD	Supply current from AVDD @2GHz		20		mA
IPD_AVDD	Power down leakage current from AVDD		1		μA

Parameter	Conditions/Comments	Min	Typical	Max	Unit
IPVDD	Supply current from PVDD @2GHz		0.1		mA
IPD_PVDD	Power down leakage current from PVDD		1		μA
IDVDD	Supply current from DVDD @2GHz		1		mA
IPD_DVDD	Power down leakage current from DVDD		1		μA
Vout Sensing Range	Sensing Range	0.675	0.75	1	V
CLK_ETC	Operating frequency		2	3	GHz
Jpp	Peak to peak period jitter			5	%
Tr	Response time			1	ns
Icom_tran	Max transient compensation current			2	A
Under_shot	Maximum value			100	mV

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Reference Documents

UIS7885 Pin List and Pin Register Instructions